

PLA/PET Recycling Lab v0.8 — 최종 BOM

Revision final-design-fabrication-closure-v0.8 · 디지털 제조 후보

구매·가공·통전 승인 문서가 아니다. donor와 supplier가 미확정인 항목을 0원 또는 승인품으로 간주하지 않는다. 실제 라벨·치수·정격·수령검사와 사용자 승인이 필요하다.

1 구성관리 기준

- Authoritative design source: bom/bom.csv (SHA-256 9b3687bc9bcb43f29b32e84c7d399cfa962f7c99d768a5a77f596d5b4ed58436)
- Subordinate detail sources: print, machine, shredder, extruder, thermal, drive manifests
- BOM rows: 156 / source design rows: 37
- 공급 상태 집계: CANDIDATE_SELECTED→RECEIPT_TEST_REQUIRED: 1, CHECK_EXISTING_STOCK_FIRST→PURCHASE_USER_APPROVAL_REQUIRED→PHYSICAL_RECEIPT_HOLD: 2, DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED: 57, DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN: 37, DIGITAL_DRAWING_PASS→FABRICATION_USER_APPROVAL_REQUIRED: 1, GATE1_QTY_1_ALLOWED_AFTER_DONOR_MEASUREMENT_AND_USER_APPROVAL: 1, NOT_APPLICABLE_REFERENCE: 6, PRINT_PACKAGE_PASS: 12, RFQ_NOT_SENT→USER_APPROVAL_REQUIRED: 2, SPECIFICATION_RELEASED→PROCUREMENT_NOT_APPROVED: 11, SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED: 12, USER_APPROVAL_AND_RECEIPT_TEST_HOLD: 4, USER_INVENTORY_VERIFICATION_REQUIRED: 4, USER_SELECTION_OR_APPROVAL_REQUIRED: 6
- Price는 정보이며 설계 release gate가 아니다. 미확정 MPN/공급사는 주문 전에 달는다.
- CSV의 17개 필드가 전체 release record이며 이 PDF는 사람이 검토하는 vector view다.

2 조달·제작 보기

Part ID	설명	수량	Make/Buy	재료/사양	승인 MPN	공급 상태
CO-01	vertical compact air cooling	1	MIXED	ABS duct and donor fans	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED→PROCUREMENT_NOT_APPROVED
CO-02	cooling fan current plus independent tach feedback	1	MIXED	calibrated A4 current feedback plus two fan tach lines through 2:1 mux to A14	NONE_APPROVED—exact make/model and receipt evidence pending	USER_SELECTION_OR_APPROVAL_REQUIRED
CT-01	Arduino Mega user interface	1	VERIFY_REUSE_OR_BUY	CT-ENC-01 sheet enclosure plus Mega TFT encoder buttons and fixed 18x18 grounded cable ducts	NONE_APPROVED—exact make/model and receipt evidence pending	USER_INVENTORY_VERIFICATION_REQUIRED
CT-ENC-01	Control-panel sheet enclosure	1	MAKE_TO_DRAWING	2 mm 5052 aluminum; process=laser + brake + PE stud	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-01	Cycloidal hook cutter disc	12	MAKE_TO_DRAWING	6 mm AISI D2 tool steel (JIS SKD11 equivalent); process=rough waterjet, vacuum harden/double temper 58-60 HRC, finish profile/faces grind	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-02	Cutter spacer	10	MAKE_TO_DRAWING	S45C normalized steel; process=turn, then match-grind/lap as one numbered two-shaft stack	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-03	Bearing side plate	2	MAKE_TO_DRAWING	12 mm S275JR steel; process=laser + bearing-seat finish	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-04	5 mm aperture screen	2	MAKE_TO_DRAWING	3 mm 304 stainless; process=laser cut + deburr	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-05	25 mm keyed cutter shaft left	1	MAKE_TO_DRAWING	S45C QT steel; process=turn + indexed keyway; black oxide with bearing seats masked	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-05R	25 mm keyed cutter shaft right	1	MAKE_TO_DRAWING	S45C QT steel; process=turn + indexed keyway; black oxide with bearing seats masked	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-06	Phase gear axial spacer	2	MAKE_TO_DRAWING	S45C normalized steel; process=simple turning	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-07	DRV-01 universal donor motor plate	1	MAKE_TO_DRAWING	6 mm steel; process=laser cut + deburr; standard metal angles	NOT_APPLICABLE—make from released file	GATE1_QTY_1_ALLOWED_AFTER_DONOR_MEASUREMENT_AND_USER_APPROVAL
CUT-08	Dual 61905 bearing retainer	2	MAKE_TO_DRAWING	2 mm steel; process=laser cut + deburr	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-09	Shredder chamber distance sleeve	4	MAKE_TO_DRAWING	S275JR steel; process=turn, drill/ream and match-face as a set	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
CUT-10	61905 bearing seat ring	4	MAKE_TO_DRAWING	S45C normalized steel; process=turn and match-face	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
DG-01	dual-axis shadow gauge	1	BUY	LED photodiode slit	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
DR-EXT	external qualified pre-dryer	1	USER_SUPPLIED_EXTERNAL	project lab/user equipment	NOT_APPLICABLE—build to released specification	USER_INVENTORY_VERIFICATION_REQUIRED
DR-GGM-01	selected GGM drive fabricated support/protection subassembly	1	MAKE_TO_DRAWING	exports/final/manufacturing/drive_ggm sub-manifest; S275/S45C/steel stock	NOT_APPLICABLE—build to released specification	DIGITAL_DRAWING_PASS→FABRICATION_USER_APPROVAL_REQUIRED
DR-GGM-02	GGM radial/jackshaft bearings	3	VERIFY_REUSE_OR_BUY	6201-2RS 12x32x10	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
DR-GGM-03	motor-lead current sensors for GGM drives	2	VERIFY_REUSE_OR_BUY	bidirectional Hall current sensor >= +/-10 A, 5 V-compatible analog output	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
DRV-03	M3 Z16 solid phase gear left	1	MAKE_TO_DRAWING	18 mm S45C; process=wire-EDM profile/bore/keyway + finish	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
DRV-03R	M3 Z16 solid phase gear right	1	MAKE_TO_DRAWING	18 mm S45C; process=wire-EDM profile/bore/keyway + finish	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
DRV-GD-01	Interlocked drive guard	1	MAKE_TO_DRAWING	1 mm galvanized steel; process=laser + brake + service-cover hardware	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-01	16 mm x 16D screw barrel and EX-DIE-01 to 05 connected open-die	1	MAKE_CNC	SCM440/304/17-4PH/C110	NOT_APPLICABLE—build to released specification	RFQ_NOT_SENT→USER_APPROVAL_REQUIRED
EX-02	thrust bearing plate clamps	1	MAKE_CNC	EX-THR-01 metal	NOT_APPLICABLE—build to released specification	RFQ_NOT_SENT→USER_APPROVAL_REQUIRED
EX-03	barrel process heater zones	3	BUY_CUSTOM	TH-BH-01 custom 24 V 100 W ID34 W45 mica band	NONE_APPROVED—exact make/model and receipt evidence pending	USER_SELECTION_OR_APPROVAL_REQUIRED
EX-04	die heater	1	BUY_CUSTOM	TH-DIE-01 Tempco custom 24 V 60 W Ø6.500 CG x39.50 flanged cartridge	NONE_APPROVED—exact make/model and receipt evidence pending	USER_SELECTION_OR_APPROVAL_REQUIRED
EX-05	thermal sensors and switching	1	MIXED	T1-T5 ungrounded mineral-insulated K-type plus MAX6675-class common-reference interfaces and 5 low-side MOSFET channels	NONE_APPROVED—exact make/model and receipt evidence pending	USER_SELECTION_OR_APPROVAL_REQUIRED
EX-06	screw shaft tach	1	MIXED	Hall sensor plus one or more shaft magnets and protected A13 input	NONE_APPROVED—exact make/model and receipt evidence pending	USER_SELECTION_OR_APPROVAL_REQUIRED
EX-07	extruder geared DC drive	1	BUY	GGM K9DG60N2 24V + K9G150C	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
EX-08	extruder motor driver/current feedback	1	VERIFY_REUSE_OR_BUY	owned BTS7960 module + external motor-lead current sensor	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
EX-BAR-01	ID16.20 x OD34 barrel	1	MAKE_TO_DRAWING	SCM440 (KS D3867/JIS G4105 equivalent) QT + gas nitride bore; process=machine integral rear shoulder from oversized blank, deep drill, stress relieve, ream/hone, port, die-interface thread and sensor bores, nitride, final hone	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-CPN-BAR	Matched barrel process coupon	1	MAKE_TO_DRAWING	same certified SCM440 heat as EX-BAR-01; process=same bore/hone/nitride route as EX-BAR-01	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED

Part ID	설명	수량	Make/Buy	재료/사양	승인 MPN	공급 상태
EX-CPN-SCR	Three-pitch screw process coupon	1	MAKE_TO_DRAWING	same certified SCM440 heat as EX-SCR-01; process=same flight mill/polish/nitride route as EX-SCR-01	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-DIE-01	Connected 90 degree down-die body	1	MAKE_TO_DRAWING	SCM440 QT + gas nitride; process=6-face mill; gun drill/ream intersecting Ø8 channels; counterbore, drill/tap; stress relieve; gas nitride; lap sealing face	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-DIE-02	Seven-hole breaker plate	1	MAKE_TO_DRAWING	304 stainless; process=wire EDM or laser + double-side lap	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-DIE-03	Replaceable Ø3 die insert	1	MAKE_TO_DRAWING	17-4PH H900 stainless; process=turn, drill/ream land, 60 degree included entrance blend, H900, finish lap	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-DIE-04	Sacrificial die relief retainer	1	MAKE_TO_DRAWING	304 stainless sheet t1.5; process=laser/waterjet + deburr; no heat treatment	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-DIE-05	Annealed copper face gasket	2	MAKE_TO_DRAWING	C110 annealed copper t0.5; process=waterjet/punch; anneal after cutting; bag clean	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-MT-01	rear hot-zone axial datum plate	1	MAKE_CNC	S275 steel plate t8	NOT_APPLICABLE—build to released specification	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
EX-MT-02	front hot-zone radial sliding guide	1	MAKE_CNC	S275 steel plate t8	NOT_APPLICABLE—build to released specification	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
EX-MT-03	rear split fixed collar	1	MAKE_CNC	S45C	NOT_APPLICABLE—build to released specification	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
EX-MT-04	rear hot-zone support rail	1	MIXED	2020 aluminum extrusion L390	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
EX-SCR-01	16 mm x 16D single screw	1	MAKE_TO_DRAWING	SCM440 (KS D3867/JIS G4105 equivalent) QT + gas nitride; process=turn between centres, 4-axis flight mill, polish, nitride, finish grind	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-SH-01	Three-panel hot-zone shield	1	MAKE_TO_DRAWING	2 mm 5052 aluminum; process=laser + two 90deg bends; bond PE	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
EX-THR-01	Extruder thrust plate	1	MAKE_TO_DRAWING	12 mm S45C normalized steel; process=laser rough + bore/seat finish	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
ExtruderFixed Collar	reference alias: EX-MT-03	1	REFERENCE_ONLY	S45C steel; process=turn, split, black oxide with bore masked	NOT_APPLICABLE	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
ExtruderFront SlidingGuide	reference alias: EX-MT-02	1	REFERENCE_ONLY	8 mm S275 steel; process=laser/waterjet rough + bore/face finish	NOT_APPLICABLE	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
ExtruderRear FixedDatum	reference alias: EX-MT-01	1	REFERENCE_ONLY	8 mm S275 steel; process=laser/waterjet rough + bore/face finish	NOT_APPLICABLE	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
ExtruderRear Retainer	reference alias: EX-MT-05	1	REFERENCE_ONLY	4 mm S355J2 steel; process=laser/waterjet rough + face and tap finish	NOT_APPLICABLE	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
ExtruderRear Retainer Spacer318	reference alias: EX-MT-06	2	REFERENCE_ONLY	S275 steel; process=turn, face and deburr; matched pair	NOT_APPLICABLE	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
Extruder SupportRail Rear	reference alias: EX-MT-04	1	REFERENCE_ONLY	2020 aluminum profile L390; process=saw cut, face and deburr	NOT_APPLICABLE	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FB-01	removable flake bin	1	MIXED	FD-BIN-01 1 mm sheet plus 4x PPR-C03	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
FD-BIN-01	Removable flake bin	1	MAKE_TO_DRAWING	1.00 ±0.05 mm 304 stainless sheet; process=laser cut, fold and rivet/TIG seam	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FD-CP-01	Feeder positive drive coupling	1	MAKE_TO_DRAWING	S45C normalized steel; process=turn + broach 3 mm keyway + cross-drill	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FD-DA-01	Feeder reference-drive frame mount	1	MAKE_TO_DRAWING	S275 steel; process=laser/waterjet + bore finish + weld flange	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FD-GSK-01	Feed-hopper flange gasket	2	MAKE_TO_DRAWING	food-contact platinum silicone 0.50±0.05 mm; process=die cut	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FD-HOP-01	Sealed feed hopper	1	BUY_TO_SPEC	2 mm 304 stainless; process=roll cone/cylinder + TIG flange + gasketed lid	NONE_APPROVED—exact make/model and receipt evidence pending	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FD-MET-01	Metering auger housing	1	MAKE_TO_DRAWING	304 stainless; process=turn tube/flanges + drill	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FD-MET-02	Positive-displacement metering auger	1	MAKE_TO_DRAWING	304 stainless; process=turn root/bore + mill or weld continuous flight	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FD-MET-03	Common auger and anti-bridge agitator shaft	1	MAKE_TO_DRAWING	304 shaft; process=turn Ø8 shaft + cross-drill + weld/pin Ø4 paddles	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FH-01	sealed steep-wall hopper with positive metering feed	1	MIXED	PF-01..PF-05 auger agitator throat and cover	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
FH-02	hopper maintenance PTC assembly	1	MIXED	4x 35x21x5 class PTC plus TH-PTC-01/02 metal spreader clamp	NONE_APPROVED—exact make/model and receipt evidence pending	CANDIDATE_SELECTED—RECEIPT_TEST_REQUIRED
FH-03	auger and agitator drive sensing interface	1	MIXED	donor motors plus independent current and tach feedback	NONE_APPROVED—exact make/model and receipt evidence pending	USER_SELECTION_OR_APPROVAL_REQUIRED
FM-AX-01	Puller roller spindle	2	MAKE_TO_DRAWING	Ø8 h6 stainless shaft; process=cut/face + collar flats	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FM-EB-01	Puller eccentric pressure bushing	2	MAKE_TO_DRAWING	S45C normalized steel; process=turn eccentric bore + mill flange slots	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FM-GA-01	Guide roller fixed axle	1	MAKE_TO_DRAWING	Ø5 h6 stainless shaft; process=cut/face + E-clip grooves or collars	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FM-GC-01	Guide roller 625 outer-ring retainer	2	MAKE_TO_DRAWING	1 mm 304 stainless sheet; process=laser cut + deburr	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FM-GR-01	Solid-strand guide roller	1	MAKE_TO_DRAWING	POM-C; process=turn + bearing-seat bore	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FM-PL-01	Puller side plate	2	MAKE_TO_DRAWING	10 mm 6061-T6; process=waterjet + ream/tap	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FM-RL-01	Puller roller	2	MAKE_TO_DRAWING	6061-T6 hub + replaceable Shore A 50–70 silicone sleeve; process=turn + bore	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
FR-01	compact profile frame cut set	1	VERIFY_REUSE_OR_BUY	20x20 aluminum profile 890x4 430x10 660x6 300x2 318x1 280x2 50x1 plus local 20x40 660x2	NONE_APPROVED—exact make/model and receipt evidence pending	USER_INVENTORY_VERIFICATION_REQUIRED
GD-01	cutter and hot-zone guards	1	MIXED	metal sheet plus ABS cold covers	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
GGM_Chain Guard	GGM drive manufactured part: GGM_ChainGuard	1	MAKE	1mm folded steel plus2x20x20 mounting tabs; twoM3 to front bearing plate; interlock required; process=1 mm 강판 절단/절곡 + 두 mounting tab	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_EX_CouplingGuard	GGM drive manufactured part: GGM_EX_CouplingGuard	1	MAKE	split metal cover: 2 formed halves, OD76x2 front flange, fourM3 mounts, 1mm seam; process=1 mm 강판 롤 + 2 mm half flange	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN

Part ID	설명	수량	Make/Buy	재료/사양	승인 MPN	공급 상태
GGM_EX_Fuse Input	GGM drive manufactured part: GGM_EX_FuseInput	1	MAKE	S45C hubs / brass uncalibrated pin blank; release target8.8-9.3Nm; process=선삭 + 키홀 + 드릴/탭	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_EX_Fuse Output	GGM drive manufactured part: GGM_EX_FuseOutput	1	MAKE	S45C hubs / brass uncalibrated pin blank; release target8.8-9.3Nm; process=선삭 + 키홀 + 드릴/탭	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_EX_Fuse PinBlank	GGM drive manufactured part: GGM_EX_FusePinBlank	1	MAKE	S45C hubs / brass uncalibrated pin blank; release target8.8-9.3Nm; process=황동봉 선삭 + 교차 드릴	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_EX_Motor Key	GGM drive manufactured part: GGM_EX_MotorKey	1	MAKE	4x4 steel key; process=규격 키 stock 절단	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_EX_Mount	GGM drive manufactured part: GGM_EX_Mount	1	MAKE	6 mm S275 plate; use catalog eccentric output datum; process=판재 절단 + 드릴	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_EX_Radial Cap	GGM drive manufactured part: GGM_EX_RadialCap	1	MAKE	steel t2; ID27.4 outer-ring-only retainer; M3 screws; process=2 mm 판재 절단 + 드릴	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_EX_Radial Holder	GGM drive manufactured part: GGM_EX_RadialHolder	1	MAKE	S275 t10; bore32.03; 4x4.5; matching M4 in thrust plate; process=판재 보링/리머 + 드릴/탭	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_EX_Screw Key	GGM drive manufactured part: GGM_EX_ScrewKey	1	MAKE	4x4 steel key; process=규격 키 stock 절단	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_JackInner Spacer_front	GGM drive manufactured part: GGM_JackInnerSpacer_front	1	MAKE	steel ID12.1 OD17 shim/spacer; inner ring only; process=강관/washer 선삭	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_JackInner Spacer_pinionfront	GGM drive manufactured part: GGM_JackInnerSpacer_pinionfront	1	MAKE	steel ID12.1 OD17 shim/spacer; inner ring only; process=강관/washer 선삭	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_JackInner Spacer_pinionrear	GGM drive manufactured part: GGM_JackInnerSpacer_pinionrear	1	MAKE	steel ID12.1 OD17 shim/spacer; inner ring only; process=강관/washer 선삭	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_JackInner Spacer_rear	GGM drive manufactured part: GGM_JackInnerSpacer_rear	1	MAKE	steel ID12.1 OD17 shim/spacer; inner ring only; process=강관/washer 선삭	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_12T	ANSI #35 12T steel keyed sprocket	1	BUY_TO_SPEC	BUY #35 12T steel, hub15, key4.1; GGM_SH_Jackshaft D12 h6; 4x4 steel; key flank carries torque	NONE_APPROVED—received keyed #35 sprocket must pass P1/P4 fit and retention evidence	CHECK_EXISTING_STOCK_FIRST—PURCHASE_USER_APPROVAL_REQUIRED—PHYSICAL_RECEIPT_HOLD
GGM_SH_30T	ANSI #35 30T steel keyed sprocket	1	BUY_TO_SPEC	BUY #35 30T steel, hub28, key6.1; CUT-05R D25 h6 drive-end land; 6x6x20 steel; key flank carries torque	NONE_APPROVED—received keyed #35 sprocket must pass P1/P4 fit and retention evidence	CHECK_EXISTING_STOCK_FIRST—PURCHASE_USER_APPROVAL_REQUIRED—PHYSICAL_RECEIPT_HOLD
GGM_SH_Angle201	GGM drive manufactured part: GGM_SH_Angle201	1	MAKE	S275 stock angle; continuous3mm fillet both legs; jig before welding; process=6 mm 판재 두 조각 + 용접	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Angle83	GGM drive manufactured part: GGM_SH_Angle83	1	MAKE	S275 stock angle; continuous3mm fillet both legs; jig before welding; process=6 mm 판재 두 조각 + 용접	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Base	GGM drive manufactured part: GGM_SH_Base	1	MAKE	6 mm S275 sheet; fourM6 base holes plus fourM5 end-tap post holes; process=판재 절단 + 드릴	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_BearingCap271	GGM drive manufactured part: GGM_SH_BearingCap271	1	MAKE	2 mm steel; M3 retainer screws; process=2 mm 판재 절단 + 드릴	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_BearingCap283	GGM drive manufactured part: GGM_SH_BearingCap283	1	MAKE	2 mm steel; M3 retainer screws; process=2 mm 판재 절단 + 드릴	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_BearingCap301	GGM drive manufactured part: GGM_SH_BearingCap301	1	MAKE	2 mm steel; M3 retainer screws; process=2 mm 판재 절단 + 드릴	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_BearingCap313	GGM drive manufactured part: GGM_SH_BearingCap313	1	MAKE	2 mm steel; M3 retainer screws; process=2 mm 판재 절단 + 드릴	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_BearingPlate273	GGM drive manufactured part: GGM_SH_BearingPlate273	1	MAKE	216x60x10 plate,32.03 bore,193x40 asymmetric hole grid; rings retained by caps; process=판재 + 보링/리머 + 탭	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_BearingPlate303	GGM drive manufactured part: GGM_SH_BearingPlate303	1	MAKE	216x60x10 plate,32.03 bore,193x40 asymmetric hole grid; rings retained by caps; process=판재 + 보링/리머 + 탭	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_CouplingGuard	GGM drive manufactured part: GGM_SH_CouplingGuard	1	MAKE	split metal cover: 2 formed halves, OD76x2 front flange, fourM3 mounts, 1mm seam; process=2 mm 강판 롤 + 분할 seam	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Cutter Key	GGM drive manufactured part: GGM_SH_CutterKey	1	MAKE	6x6x20 steel key; process=규격 키 stock 절단	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Fuse Input	GGM drive manufactured part: GGM_SH_FuseInput	1	MAKE	S45C hubs / brass uncalibrated pin blank; release target8.8-9.3Nm; process=선삭 + 키홀 + 드릴/탭	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Fuse Output	GGM drive manufactured part: GGM_SH_FuseOutput	1	MAKE	S45C hubs / brass uncalibrated pin blank; release target8.8-9.3Nm; process=선삭 + 키홀 + 드릴/탭	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Fuse PinBlank	GGM drive manufactured part: GGM_SH_FusePinBlank	1	MAKE	S45C hubs / brass uncalibrated pin blank; release target8.8-9.3Nm; process=황동봉 선삭 + 교차 드릴	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Jack InputKey	GGM drive manufactured part: GGM_SH_JackInputKey	1	MAKE	4x4 steel key; process=규격 키 stock 절단	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Jack SprocketKey	GGM drive manufactured part: GGM_SH_JackSprocketKey	1	MAKE	4x4 steel key; process=규격 키 stock 절단	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Jackshaft	GGM drive manufactured part: GGM_SH_Jackshaft	1	MAKE	S45C 12 h6 x78; two 4 mm keys; process=규격 S45C 봉 + 선삭 + 키홀	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Motor Key	GGM drive manufactured part: GGM_SH_MotorKey	1	MAKE	4x4 steel key; process=규격 키 stock 절단	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Mount	GGM drive manufactured part: GGM_SH_Mount	1	MAKE	6 mm S275 plate; 4x6.6 PCD104; pilot36.5; process=판재 절단 + 드릴 + 지그 용접	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Spacer_220_100	GGM drive manufactured part: GGM_SH_Spacer_220_100	1	MAKE	steel spacer 12/6.6 x10; process=규격 강관 절단 + 끝면 다듬기	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Spacer_220_280	GGM drive manufactured part: GGM_SH_Spacer_220_280	1	MAKE	steel spacer 12/6.6 x10; process=규격 강관 절단 + 끝면 다듬기	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Spacer_90_100	GGM drive manufactured part: GGM_SH_Spacer_90_100	1	MAKE	steel spacer 12/6.6 x10; process=규격 강관 절단 + 끝면 다듬기	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
GGM_SH_Spacer_90_280	GGM drive manufactured part: GGM_SH_Spacer_90_280	1	MAKE	steel spacer 12/6.6 x10; process=규격 강관 절단 + 끝면 다듬기	NOT_APPLICABLE_CUSTOM	DIGITAL_DRAWING_PASS; PHYSICAL_NOT_RUN
HP-01	refillable sliding-lid hopper	1	MIXED	IN-HOP-01 plus PPR-C01/C02	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
IN-HOP-01	Refillable input hopper	1	MAKE_TO_DRAWING	2 mm 5052-H32 aluminum; process=roll cone/cylinder + TIG weld + deburr	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
PL-01	puller pair with tach	1	MIXED	FM-PL-01/RL-01/AX-01 plus donor motor and tach	NONE_APPROVED—exact make/model and receipt evidence pending	USER_INVENTORY_VERIFICATION_REQUIRED
PPR-C01	Sliding hopper lid	1	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.24 mm; 4 walls; 20% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS

Part ID	설명	수량	Make/Buy	재료/사양	승인 MPN	공급 상태
PPR-C02	Anti-reach baffle chute	1	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.24 mm; 5 walls; 25% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C03	Flake bin sheet corner	4	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.28 mm; 4 walls; 25% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C04	Screen drawer handle	1	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.24 mm; 5 walls; 35% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C05	Cooling duct segment	2	MAKE_3D_PRINT	ABS; process=FDM; 0.4 mm nozzle; 0.24 mm; 4 walls; 15% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C06	Gauge enclosure	2	MAKE_3D_PRINT	ABS; process=FDM; 0.4 mm nozzle; 0.20 mm; 4 walls; 25% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C07	Puller pinch guard	1	MAKE_3D_PRINT	ABS; process=FDM; 0.4 mm nozzle; 0.24 mm; 5 walls; 20% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C08	Solid-strand guide axle bracket	2	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.20 mm; 5 walls; 40% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C09	Spool cone adapter	2	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.20 mm; 5 walls; 35% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C10	Traverse carriage	1	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.20 mm; 5 walls; 40% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C11	Control panel bezel	1	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.20 mm; 4 walls; 20% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-C12	Cable duct clamp	8	MAKE_3D_PRINT	PLA; process=FDM; 0.4 mm nozzle; 0.20 mm; 4 walls; 50% infill	NOT_APPLICABLE—make from released file	PRINT_PACKAGE_PASS
PPR-EXTRUDER-ASM	reference alias: PPR-EXTRUDER-ASM	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	NOT_APPLICABLE_REFERENCE
PPR-FEEDER-ASM	reference alias: PPR-FEEDER-ASM	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	NOT_APPLICABLE_REFERENCE
PPR-FORMING-ASM	reference alias: PPR-FORMING-ASM	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	NOT_APPLICABLE_REFERENCE
PPR-FRAME-ASM	reference alias: PPR-FRAME-ASM	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	NOT_APPLICABLE_REFERENCE
PPR-FULL-ASM	reference alias: PPR-FULL-ASM	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	NOT_APPLICABLE_REFERENCE
PPR-SHREDDER-ASM	reference alias: PPR-SHREDDER-ASM	1	REFERENCE_ONLY	not separately procured; canonical BOM item or assembly	NOT_APPLICABLE	NOT_APPLICABLE_REFERENCE
SF-01	latching E-stop and hardwired cut	1	BUY	qualified switch plus relay topology	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
SF-02	lid and service interlocks	3	BUY	positive opening switch candidate	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
SF-03	thermal and branch fuses	1	BUY	temperature/current rated	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
SH-01	dual-shaft hook cutter assembly	1	MAKE_CNC	steel	NOT_APPLICABLE—build to released specification	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
SH-02	removable 5 mm screen	1	MAKE_CNC	stainless steel	NOT_APPLICABLE—build to released specification	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
SH-03	interchangeable shredder geared DC drive	1	BUY	GGM K9DG60N2 24V + K9G75C + GGM drive r2 support/jackshaft; existing #35 12T:30T	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
SH-04	interchangeable chain and phase drive	1	MIXED	direct-keyed GGM_SH_12T/GGM_SH_30T #35 sprockets + DRV-03/03R phase pair + GGM 12 mm jackshaft chain path	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
SH-05	bidirectional motor driver/current feedback	1	VERIFY_REUSE_OR_BUY	owned BTS7960 module + external motor-lead current sensor	NONE_APPROVED—exact make/model and receipt evidence pending	SUPPLIER_OR_INVENTORY_VERIFICATION_REQUIRED
SP-01	dancer traverse spooler	1	MIXED	FM-GR/GA SP-DA/AX/RL/SH plus donor motion tach left/right limits and PPR-C08..10	NONE_APPROVED—exact make/model and receipt evidence pending	SPECIFICATION_RELEASED—PROCUREMENT_NOT_APPROVED
SP-AX-01	Dancer pivot/roller axles	2	MAKE_TO_DRAWING	Ø8 h6 stainless shaft; process=cut/face + collars	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-BP-01	Spool 6001 bearing pocket plate	2	MAKE_TO_DRAWING	10 mm 6061-T6; process=waterjet rough + pocket bore finish	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-BR-01	Spool 6001 outer-ring retainer	2	MAKE_TO_DRAWING	2 mm 304 stainless sheet; process=laser cut + deburr	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-DA-01	Dancer arm	1	MAKE_TO_DRAWING	8 mm 6061-T6; process=waterjet + ream	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-DS-01	Dancer pivot support plate	1	MAKE_TO_DRAWING	8 mm 6061-T6; process=waterjet + ream	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-MM-01	Universal NEMA17-class spool motor plate	1	MAKE_TO_DRAWING	6 mm 6061-T6; process=laser/waterjet + drill	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-RL-01	Dancer end roller	1	MAKE_TO_DRAWING	POM-C; process=turn + bore	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-SH-01	Spool spindle	1	MAKE_TO_DRAWING	Ø12 h6 S45C; process=cut/turn faces + collar flats	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
SP-TR-01	Traverse rod end plate	2	MAKE_TO_DRAWING	5 mm 6061-T6; process=waterjet + ream	NOT_APPLICABLE—make from released file	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
TH-BH-01	Custom barrel mica band heater	3	BUY_TO_SPEC	mica/NiCr/stainless sheath; process=custom heater RFQ	NONE_APPROVED—exact make/model and receipt evidence pending	DIGITAL_DRAWING_PASS; PART_GATE=PASS; PROCUREMENT_USER_APPROVAL_REQUIRED
TH-DIE-01	Die cartridge heater	1	BUY_TO_SPEC	Tempco custom Hi-Density; 321SS sheath; 304SS flange; process=buy/custom fabricate to thermal drawing note	NONE_APPROVED—exact make/model and receipt evidence pending	USER_APPROVAL_AND_RECEIPT_TEST_HOLD
TH-FUSE-01	Independent one-shot thermal fuse envelope	3	BUY_TO_SPEC	300 C-class barrel/die protection; process=buy/custom fabricate to thermal drawing note	NONE_APPROVED—exact make/model and receipt evidence pending	USER_APPROVAL_AND_RECEIPT_TEST_HOLD
TH-TC-01	Tempco MTA1 custom ungrounded Type-K probes	5	BUY_TO_SPEC	Alloy 600 MI sheath Ø3.00 ±0.03; 96% MgO; supplier-welded 304SS stop collar; process=buy/custom fabricate to thermal drawing note	NONE_APPROVED—exact make/model and receipt evidence pending	USER_APPROVAL_AND_RECEIPT_TEST_HOLD
TH-TCR-01	Thermocouple stop-collar retainer bridge	4	BUY_TO_SPEC	304 stainless t1.5; process=buy/custom fabricate to thermal drawing note	NONE_APPROVED—exact make/model and receipt evidence pending	USER_APPROVAL_AND_RECEIPT_TEST_HOLD

3 인터페이스·도면·조립 교차참조

Part ID	핵심 인터페이스	도면	조립 단계	펌웨어 의존	비고
CO-01	hot-shield clearance and verified airflow feedback	docs/drawings/v0.8/FM-001_cooling_strand_path.svg	docs/final/complete_build_manual_ko.pdf step 14; docs/final/assembly_steps.csv	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=DESIGNED; quantity_unit=each; two serviceable duct segments
CO-02	hot-shield clearance and verified airflow feedback	docs/drawings/v0.8/FM-001_cooling_strand_path.svg	docs/final/complete_build_manual_ko.pdf step 14; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=USER_APPROVAL_REQUIRED; quantity_unit=each; commanded PWM with 1.5 s bounded feedback dwell; each fan stopped/both stopped/implausible-off detection; tach is not airflow
CT-01	PE bond, segregated routing and pin-map consistency	docs/drawings/v0.8/EL-001_electrical_enclosure.svg	docs/final/complete_build_manual_ko.pdf step 19; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=project lab; cash_class=REUSE_PENDING; status=UNVERIFIED; quantity_unit=each; profile lock and cleaning wizard; heater and sensor conductors require partitioned routing; duct stock receipt measurement pending
CT-ENC-01	PE bond, segregated routing and pin-map consistency; drawing tolerance=190 x35 x190; service-open face; PPR-C11 bezel datum; M4 profile mounts; segregate heater/ motor and signal wiring; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CT-ENC-01.pdf	docs/final/complete_build_manual_ko.pdf step 19; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-01	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=OD 58.0; root circle R18.0 with smooth DXF transition; bore 25.01 +0.01/0; matched keyway width 6.005-6.010; final profile ±0.10 and taper ≤0.05 through thickness after CAM kerf compensation; t6.00 ±0.03; flatness/parallelism 0.03; tooth side deburr C0.15 max; axial working gap 0.25-0.50 mm by metal shim; selected 5.995-6.000 key plus engraved index line registers stack; replace on crack, chip >0.5 mm, edge recession >0.30 mm or lost shim gap; datum=A=shaft bore axis; B=ground stack face; C=keyway centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-01.pdf	docs/final/complete_build_manual_ko.pdf step 6; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-02	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=OD34.0; bore25.10 +0.05/0; nominal length7.00; faces parallel within0.01; engrave shaft/ position; with CUT-01 and 0.05/0.10/0.25 metal shims, every one of 11 opposing axial gaps shall measure0.25-0.50 over one full hand rotation; supply only as the accepted matched set; datum=A=shaft bore axis; B=ground stack face; C=keyway centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-02.pdf	docs/final/complete_build_manual_ko.pdf step 6; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-03	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=two SKF 61905-2RS1 seats diameter 42 H7; center distance 48.00 +/-0.03; match-machine both plates; seat-axis parallelism 0.05/140; four frame holes diameter 6.6; identify front fixed and rear floating sides; after CUT-10 retention and metal shimming, each complete shaft axial float shall be0.05-0.20 over one full hand rotation; datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-03.pdf	docs/final/complete_build_manual_ko.pdf step 4; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-04	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=135 x 120 x 3; apertures diameter 5.0 on 9.0 pitch; all strand-side edges R0.3; verify minimum 1.9 mm rotating clearance with shims before powered test; datum=A=shaft bore axis; B=ground stack face; C=keyway centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-04.pdf	docs/final/complete_build_manual_ko.pdf step 9; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-05	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=continuous Ø25 h6 =24.987-25.000 across 61905 journals, cutter land and gear/hub land; overall240.0 +/-0.10; TIR≤0.05; 6.005-6.010 keyways at y=0-35 and55-160 with selected key5.995-6.000; 8.005-8.010 keyway at195-240 with selected key7.995-8.000; datum clock0 deg; install slave shaft at Y278; datum=A=common journal axis; B=drive-end shoulder/ face; C=keyway or retaining feature plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-05.pdf	docs/final/complete_build_manual_ko.pdf step 5; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-05R	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=continuous Ø25 h6 =24.987-25.000 across 61905 journals, cutter land and gear/hub land; overall240.0 +/-0.10; TIR≤0.05; all keyway centreplanes clocked25.714 +/-0.02 deg; 6.005-6.010 keyways at y=0-35 and55-160 with selected key5.995-6.000; 8.005-8.010 keyway at195-240 with selected key7.995-8.000; install driven shaft at Y258; datum=A=common journal axis; B=drive-end shoulder/face; C=keyway or retaining feature plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-05R.pdf	docs/final/complete_build_manual_ko.pdf step 5; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-06	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=OD 34.0; bore 25.10 +0.05/0; length 4.00 +/-0.03; faces parallel within 0.03; datum=A=shaft bore axis; B=stack face; C=key/dowel angular datum; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-06.pdf	docs/final/complete_build_manual_ko.pdf step 6; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-07	shaft/bearing fit; cutter shim clearance; phase registration	exports/cnc/CUT-07/drawing_notes.md	docs/final/complete_build_manual_ko.pdf step 1; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
CUT-08	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=figure-eight OD lobes 60; two relief bores diameter 34; center distance 48.00 +/-0.05; six M4 clearance holes diameter 4.5 at drawing coordinates; CUT-03 matching holes are included and may be match-reamed	exports/final/manufacturing/RFQ/CUT-08.pdf	docs/final/complete_build_manual_ko.pdf step 4; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv

Part ID	핵심 인터페이스	도면	조립 단계	필웨어 의존	비고
	after bearing-seat finish; datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report				
CUT-09	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=OD10.00 +/-0.05; ID6.60 +0.10/0; length128.00 +/-0.03; four-piece matched length spread <=0.03; face parallelism <=0.03; deburr C0.2 max; datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-09.pdf	docs/final/complete_build_manual_ko.pdf step 1; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
CUT-10	shaft/bearing fit; cutter shim clearance; phase registration; drawing tolerance=OD42 g6; ID25.50 +0.10/0; width3.00 +/-0.02; faces parallel0.01; bears on outer ring only; engrave front-fixed/rear-floating and shaft ID; set rear outer-ring endplay using0.05/0.10/0.20 ground metal shims; complete-shaft axial float acceptance0.05-0.20 mm; datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/CUT-10.pdf	docs/final/complete_build_manual_ko.pdf step 1; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/shredder_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
DG-01	orthogonal gauge alignment and calibration	docs/drawings/v0.8/FM-002_gauge_puller.svg	docs/final/complete_build_manual_ko.pdf step 15; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=stock/buy; cash_class=MIXED; status=CALIBRATION_HOLD; quantity_unit=each; no accuracy claim
DR-EXT	external dryer qualification and material moisture evidence	docs/drawings/v0.8/GA-001_general_arrangement.svg	docs/final/complete_build_manual_ko.pdf step 1; docs/final/assembly_steps.csv	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=external; cash_class=EXCLUDED; status=UNVERIFIED; quantity_unit=each; not inside machine envelope or cash budget
DR-GGM-01	selected GGM drive mounting, bearing support, coupling protection and current feedback	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=project lab fabricate; cash_class=REUSE_PENDING; status=DESIGNED; quantity_unit=each; authoritative sub-manifest covers 39 drive manufacturing objects; legacy DRV-01/DRV-A60/DRV-F01A/B/P superseded
DR-GGM-02	selected GGM drive mounting, bearing support, coupling protection and current feedback	exports/final/manufacturing/drive_ggm/PPR_GGM_ASSEMBLY_INSPECTION_KO_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=check project lab stock then buy; cash_class=MIXED; status=STOCK_CHECK_REQUIRED; quantity_unit=each; 2 jackshaft + 1 extruder rear radial bearing; caps contact outer ring only; 0.05-0.20 mm axial endplay after receipt
DR-GGM-03	selected GGM drive mounting, bearing support, coupling protection and current feedback	exports/final/electrical/full_wiring_diagram.pdf	docs/final/complete_build_manual_ko.pdf step 19; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=check project lab stock then buy; cash_class=MIXED; status=STOCK_CHECK_REQUIRED; quantity_unit=each; A0 shredder/A9 extruder; independent reference calibration <=0.10 A bound and torque holdout <=0.4 N.m
DRV-03	keyed phase/sprocket interface; chain alignment; GGM protection coupling governs mechanical release; drawing tolerance=M3 Z16 20 degree; face18; assembled pair tangential backlash 0.120-0.140 over one full mesh rotation at 20 +/-2 C; bore Ø25.01 +0.01/0; matched 8.005-8.010 keyway at datum0 deg; root fillet R1 minimum; 2xM4 clearance + 1xØ3 H7 inspection/clocking holes; full34 N.m CalculiX screen; datum=A=shaft bore axis; B=stack face; C=key/dowel angular datum; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/DRV-03.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/drive_interface/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
DRV-03R	keyed phase/sprocket interface; chain alignment; GGM protection coupling governs mechanical release; drawing tolerance=same tooth form and full-rotation pair backlash acceptance as DRV-03; matched 8.005-8.010 keyway clocked14.464 +/-0.02 deg from tooth datum so installed tooth phase11.25 deg matches CUT-05R key datum25.714 deg; datum=A=shaft bore axis; B=stack face; C=key/dowel angular datum; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/DRV-03R.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/drive_interface/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
DRV-GD-01	keyed phase/sprocket interface; chain alignment; GGM protection coupling governs mechanical release; drawing tolerance=165 x48 x190; two Ø32 shaft clearances at X20/68,Z55; installed annular opening≤6; open service face; positive-opening interlock flag; PE bond; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/DRV-GD-01.pdf	docs/final/complete_build_manual_ko.pdf step 8; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
EX-01	screw/barrel cold clearance; die seal; metal thrust path	docs/drawings/v0.8/EX-002_screw_barrel_die.svg	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=CNC; cash_class=CASH; status=RFQ_HOLD; quantity_unit=each; common PLA/PET hot path; physical process/hot-relief coupons before full order
EX-02	screw/barrel cold clearance; die seal; metal thrust path	docs/drawings/v0.8/EX-002_screw_barrel_die.svg	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=CNC; cash_class=CASH; status=RFQ_HOLD; quantity_unit=each; screw to profile load path; Ø17.2 passage Ø30.2 seat
EX-03	screw/barrel cold clearance; die seal; metal thrust path	docs/drawings/v0.8/EX-002_screw_barrel_die.svg	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=custom RFQ; cash_class=CASH; status=USER_APPROVAL_HOLD; quantity_unit=each; 300 W total; stock Ø35 substitution prohibited; F-H1..H3 branch fuses mandatory; TF-BARREL/TF-DIE act in K0 coil safety chain; vendor candidates pinned in bom/procurement_v0.5_ko.md (한국전자전열 heater1.co.kr, 한일전열 han-il.com, Ali Express custom fallback)
EX-04	screw/barrel cold clearance; die seal; metal thrust path	docs/drawings/v0.8/EX-002_screw_barrel_die.svg	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=custom RFQ; cash_class=CASH; status=USER_APPROVAL_HOLD; quantity_unit=each; Ø6.55 H7 bore=6.550-6.565 mm; heater OD=6.487-6.513 mm; insertion 39.50±0.20 mm; HTL550 C leads + MFR flange; cold resistance 9.12-10.56 ohm at 20±2 C; accepted supplier drawing, insulation/hipot and full hand-fit receipt evidence required before energization
EX-05	screw/barrel cold clearance; die seal; metal thrust path	docs/drawings/v0.8/EX-002_screw_barrel_die.svg	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=USER_APPROVAL_REQUIRED; quantity_unit=each; each channel branch fuse; sheath-to-junction insulation receipt test; MAX6675 T-common electronics reference; software is not sole cutoff; vendor-pinned modules are purchase candidates only
EX-06	screw/barrel cold clearance; die seal; metal thrust path	docs/drawings/v0.8/EX-002_screw_barrel_die.svg	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=USER_APPROVAL_REQUIRED; quantity_unit=each; actual

Part ID	핵심 인터페이스	도면	조립 단계	펌웨어 의존	비고
					RPM/revolutions gate purge; mass per revolution is not calibrated and purge mass is not claimed
EX-07	screw/barrel cold clearance; die seal; metal thrust path	docs/drawings/v0.8/EX-002_screw_barrel_die.svg	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=buy; cash_class=CASH; status=SELECTED_PURCHASE_HOLD; quantity_unit=each; direct screw drive nominal20 rpm; 8.0 N.m software gearbox torque limit; 6201 radial support + existing 51102 thrust path isolate motor from extrusion thrust; reverse prohibited
EX-08	screw/barrel cold clearance; die seal; metal thrust path	docs/drawings/v0.8/EX-002_screw_barrel_die.svg	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=project lab + sensor stock-check/buy; cash_class=MIXED; status=REUSE_CONFIRMED_SENSOR_HOLD; quantity_unit=each; BTS7960 qty1 of owned qty2 allocated to screw; D6 RPWM/D33 LPWM/D34 enable; A9 current feedback; calibration NOT_RUN
EX-BAR-01	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=L280; ID16.20 +0.02/0 honed; OD34 - 0.03/0 ground; rear shoulder Ø44 -0.05/0 x8 +/-0.02, feed flat X16; face runout0.03 to Datum D; 4xM4-6H depth8 PCD26; 2xØ3 H7 locating holes PCD26 at0/180 deg, position Ø0.02 to D, selected Ø3 m6 dowels; 3xØ3.20 +0.05/0 flat-bottom blind5.40 +/-0.05 at B+95/170/245; 2x M3-6H depth4 at axial pitch10 around each sensor; sensor position0.05 to D, perpendicularity0.10/5.5; conservative ligament3.345 >=3.32; thread ligament outer/inner>=2.0/ 2.9; B rear face/C front face/D bore axis; HOLD material/ process; SCM440 QT 28~32 HRC; gas nitride 0.30~0.50 mm, surface ≥900 HV; final hone bore Ra 0.4~0.8 µm; three-station bore report; datum=B=rear face; C=front face; D=bore axis; nominal +X=feed, +Y=sensors; die holes 45+90n deg from +X; angle tolerance HOLD; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/EX-BAR-01.pdf	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
EX-CPN-BAR	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=L60.00; ID/OD/finish/case same as barrel; coupon RFQ only; datum=A=common journal/bore axis; B=drive/feed datum face; C=feed-port angular plane; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/EX-CPN-BAR.pdf	docs/final/complete_build_manual_ko.pdf step 1; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
EX-CPN-SCR	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=L48.00; three pitches; OD/root/land/ finish/case same as feed zone; coupon RFQ only; datum=A=common journal/bore axis; B=drive/feed datum face; C=feed-port angular plane; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/EX-CPN-SCR.pdf	docs/final/complete_build_manual_ko.pdf step 1; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
EX-DIE-01	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=40x40x48; barrel face X40; Ø8 melt turn; Ø16.20 +0.05/0 x3 breaker seat; Ø12.00 +0.03/0 x14 insert seat; 4xØ4.5 + Ø8 recess PCD26; 2xØ3 H7 depth6 PCD26 at0/180 deg, position Ø0.02 to melt axis, selected Ø3 m6 dowels; heater Ø6.55 H7 through; 2xM3-6H depth6 at14 mm pitch for heater flange; sensor Ø3.20 +0.05/0 blind12.00 +/-0.05; 2xM3-6H depth4 at10 mm pitch around sensor; face flatness0.03; deburr flow intersection; HOLD; datum=A=barrel interface face; B=melt-channel axis; C=fastener-pattern centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/EX-DIE-01.pdf	docs/final/complete_build_manual_ko.pdf step 13; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
EX-DIE-02	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=Ø15.90 -0.05/0 x2.00 ±0.03; 7x Ø2.00 +0.05/0, six on PCD10; flatness 0.03; all flow edges R0.15 max; HOLD with die body; datum=A=barrel interface face; B=melt-channel axis; C=fastener-pattern centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/EX-DIE-02.pdf	docs/final/complete_build_manual_ko.pdf step 13; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
EX-DIE-03	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=OD Ø11.90 -0.02/0 x14.00 ±0.03; outlet Ø3.00 +0.02/0 x10.00 land; 4 mm transition from Ø8 to Ø3; land Ra<=0.4 µm; concentricity 0.02 to OD; datum=A=barrel interface face; B=melt-channel axis; C=fastener-pattern centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/EX-DIE-03.pdf	docs/final/complete_build_manual_ko.pdf step 13; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
EX-DIE-04	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=32 x20 x1.5; two 10 wide x2.5 long bending webs; 2x Ø4.5 at 24 centres; centre bypass Ø4; flatness 0.15; coupon-calibrate at operating temperature, analytical estimate is not release evidence; datum=A=barrel interface face; B=melt-channel axis; C=fastener-pattern centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/EX-DIE-04.pdf	docs/final/complete_build_manual_ko.pdf step 13; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
EX-DIE-05	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=OD34; ID16.20 +0.10/0; 4x Ø4.5 PCD26 at45 degree; 2x Ø3.20 dowel clearance at PCD26 angles0/ 180; t0.50 ±0.03; burr <=0.03; one spare required; datum=A=barrel interface face; B=melt-channel axis; C=fastener-pattern centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/EX-DIE-05.pdf	docs/final/complete_build_manual_ko.pdf step 13; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
EX-MT-01	rear axial datum/front sliding thermal expansion path	exports/final/manufacturing/hot_zone/ExtruderRear FixedDatum.svg	docs/final/complete_build_manual_ko.pdf step 11; docs/final/assembly_steps.csv	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=CNC; cash_class=CASH; status=DESIGNED; quantity_unit=each; v0.8 STEP/DXF/PDF; bore Ø34.25 +0.05/0; procurement USER_APPROVAL_REQUIRED
EX-MT-02	rear axial datum/front sliding thermal expansion path	exports/final/manufacturing/hot_zone/Extruder FrontSlidingGuide.svg	docs/final/complete_build_manual_ko.pdf step 11; docs/final/assembly_steps.csv	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=CNC; cash_class=CASH; status=DESIGNED; quantity_unit=each; v0.8 STEP/DXF/PDF; bore Ø34.60 +0.10/0; cold axial travel at least 1.50 mm
EX-MT-03	rear axial datum/front sliding thermal expansion path	exports/final/manufacturing/hot_zone/Extruder FixedCollar.svg	docs/final/complete_build_manual_ko.pdf step 11; docs/final/assembly_steps.csv	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=CNC; cash_class=CASH; status=DESIGNED; quantity_unit=each; v0.8 STEP/DXF/PDF; bore Ø34.25 +0.03/0; front guide installation prohibited

Part ID	핵심 인터페이스	도면	조립 단계	평웨어 의존	비고
EX-MT-04	rear axial datum/front sliding thermal expansion path	exports/final/manufacturing/hot_zone/ExtruderSupportRailRear.svg	docs/final/complete_build_manual_ko.pdf step 11; docs/final/assembly_steps.csv	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=stock/cut; cash_class=CASH; status=DESIGNED; quantity_unit=each; v0.8 STEP/DXF/PDF; metal face-joint; cut length 390.0 ±0.5 mm
EX-SCR-01	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=total 316.00; active 256.00; OD 15.92 - 0.02/0; RH pitch 16.00; land 1.60; Datum A axis from Ø12 h6 drive and Ø15 h6 x11 thrust-bearing seat; integral Ø23.00 +0.05/0 x9 shaft-washer abutment, face runout0.03 to A material/process; SCM440 QT 28-32 HRC; gas nitride 0.30-0.50 mm, surface 900-1100 HV; flight OD TIR ≤0.05/256; OD Ra≤0.8 µm; datum=A=common journal/bore axis; B=drive/feed datum face; C=feed-port angular plane; verify datum marking on supplier inspection report; GGM delta: Screw-GGM.step 중 후방60 mm를 확대. 용융부 flight는 기존 EX-SCR-01 치수-재료-열처리 도면을 유지.; 구동부 D12 h6 x35. 축방향 keyseat 폭4.05 +0.03/0 x20.00 +0.05/0, end에서1~21 mm.; 베어링 land는 end에서24~34 mm: 중단 없는 원통면, Ra0.8, TIR0.03. 횡방향 notch 가공 금지.; 바닥 평면은 중심에서 3.50 mm. 기존 thrust shoulder/51102 seat를 변경하지 않는다.; 이 변경도만으로 전체 screw를 새로 발주하지 않는다. 기존 active flight source와 함께 사용.	exports/final/manufacturing/Rfq/EX-SCR-01.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/cnc/extruder/rfq_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
EX-SH-01	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=335 x75 x85; open bottom/ends; feeder opening Ø50 at X314/Y37; >=10 mm ABS-duct gap; edge hem/deburr; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/EX-SH-01.pdf	docs/final/complete_build_manual_ko.pdf step 13; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
EX-THR-01	screw/barrel cold clearance; die seal; metal thrust path; drawing tolerance=12 x95 x105; passage Ø17.2; 51102 pocket Ø28.30 +0.05/0 x9.10 +0.05/0 from marked front face; 4xØ6.6; pocket shoulder square0.05 to bore; NSK general-purpose housing radial clearance >0.25; metal-to-profile load path; datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report; GGM delta: 두께12.00 +0.10/0; 기존EX-THR-01와51102 하우징 pocket를 유지. 4xM4x0.7: 반경 holder hole grid38 x38.; R2: 후면6201 접촉 중앙에D27.40 +0.10/0 x0.30 +0.05/0 relief. 외륜의 환형 좌면만 남긴다.; 추력 bearing의 정면 pocket는 기존 D28.30~28.35 x9.10~9.15 기준이며 R2 rear relief와 혼동 금지.; 4개의 원래 M6 frame hole를 유지. bearing seat와 후면에 버/돌기 금지; 겹친 tap찾아-잔여백 확인.	exports/final/manufacturing/Rfq/EX-THR-01.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
ExtruderFixed Collar	reference identity must resolve to canonical BOM/drawing; drawing tolerance=OD Ø50; L12.00 ±0.05; bore Ø34.25 +0.03/0; datum face runout 0.05 to bore axis; bore Ra≤1.6 µm; non-contact radial locator only; no blue-fit clamp on hot barrel; datum=A=profile mounting plane; B=barrel axis; C=rear axial datum; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/ExtruderFixedCollar.pdf	docs/final/complete_build_manual_ko.pdf step 11; docs/final/assembly_steps.csv	NONE	active_part_set reference; canonical=EX-MT-03; exclude from procurement roll-up; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
ExtruderFront SlidingGuide	reference identity must resolve to canonical BOM/drawing; drawing tolerance=guide bore Ø34.60 +0.10/0; 2×Ø6.6 rail holes; datum face flatness 0.10; bore position ±0.10; cold axial travel ≥1.50 mm; no axial clamp; datum=A=profile mounting plane; B=barrel axis; C=rear axial datum; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/ExtruderFrontSlidingGuide.pdf	docs/final/complete_build_manual_ko.pdf step 11; docs/final/assembly_steps.csv	NONE	active_part_set reference; canonical=EX-MT-02; exclude from procurement roll-up; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
ExtruderRear FixedDatum	reference identity must resolve to canonical BOM/drawing; drawing tolerance=barrel bore Ø34.25 +0.05/0; 2×Ø6.6 rail holes; datum face flatness 0.10; bore axis perpendicularity 0.10/54; hole position ±0.10; datum=A=profile mounting plane; B=barrel axis; C=rear axial datum; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/ExtruderRearFixedDatum.pdf	docs/final/complete_build_manual_ko.pdf step 11; docs/final/assembly_steps.csv	NONE	active_part_set reference; canonical=EX-MT-01; exclude from procurement roll-up; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
ExtruderRear Retainer	reference identity must resolve to canonical BOM/drawing; drawing tolerance=web 4.00 ±0.05; rear ribs 8.00 ±0.05; 2×M4×0.7-6H full engagement ≥8.0; face flatness 0.03; parent yield ≥177.5 MPa at service temperature; cold endplay 0.12-0.28 mm; datum=A=profile mounting plane; B=barrel axis; C=rear axial datum; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/ExtruderRearRetainer.pdf	docs/final/complete_build_manual_ko.pdf step 11; docs/final/assembly_steps.csv	NONE	active_part_set reference; canonical=EX-MT-05; exclude from procurement roll-up; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
ExtruderRear Retainer Spacer318	reference identity must resolve to canonical BOM/drawing; drawing tolerance=OD Ø8.00 ±0.03; ID Ø4.50 +0.10/0; L4.20 ±0.02; face flatness 0.01; matched length spread ≤0.02; datum=A=profile mounting plane; B=barrel axis; C=rear axial datum; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/ExtruderRearRetainerSpacer318.pdf	docs/final/complete_build_manual_ko.pdf step 11; docs/final/assembly_steps.csv	NONE	active_part_set reference; canonical=EX-MT-06; exclude from procurement roll-up; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
ExtruderSupportRail Rear	reference identity must resolve to canonical BOM/drawing; drawing tolerance=length 390.0 ±0.5 mm; end squareness 0.3; front/rear support height difference ≤0.20 mm; axis parallelism ≤0.20/390; datum=A=profile mounting plane; B=barrel axis; C=rear axial datum; verify datum marking on supplier inspection report	exports/final/manufacturing/Rfq/ExtruderSupportRailRear.pdf	docs/final/complete_build_manual_ko.pdf step 11; docs/final/assembly_steps.csv	NONE	active_part_set reference; canonical=EX-MT-04; exclude from procurement roll-up; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv
FB-01	screen/bin clearance and service withdrawal	docs/drawings/v0.8/FD-002_recirculation_screen.svg	docs/final/complete_build_manual_ko.pdf step 9; docs/final/assembly_steps.csv	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=DESIGNED; quantity_unit=each; real 1.4 mm sheet channels and matching panel corner reliefs
FD-BIN-01	sealed flake path; feeder-to-barrel interface; drawing tolerance=185 x175 x115 outside; sheet1.00 ±0.05; PPR-C03 slot1.40 ±0.30 gives clearance0.05-0.75; corner reliefs control; no inward burr/dead pocket; removable without cutter disassembly; datum=A=material-flow axis;	exports/final/manufacturing/Rfq/FD-BIN-01.pdf	docs/final/complete_build_manual_ko.pdf step 10; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/Rfq/manifest.csv

Part ID	핵심 인터페이스	도면	조립 단계	첨웨어 의존	비고
	B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report				
FD-CP-01	sealed flake path; feeder-to-barrel interface; drawing tolerance=OD18 x24; both bores Ø8.05 +0.03/0; gearbox half 3.10 +0.05/0 keyway x12; feeder half Ø3.00 +0.05/0 cross-hole at Z6 matched to FD-MET-03; new Ø3x18 spring pin; no set-screw-only torque path; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FD-CP-01.pdf	docs/final/complete_build_manual_ko.pdf step 10; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FD-DA-01	sealed flake path; feeder-to-barrel interface; drawing tolerance=126x70x70 welded shelf; gearbox pilot Ø26 clearance; 4xØ4.5 at31 mm square; 2xØ6.6 frame holes at40 pitch; mount centre30/35; weld continuous both sides; gearbox axis position0.10 to frame datum; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FD-DA-01.pdf	docs/final/complete_build_manual_ko.pdf step 10; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FD-GSK-01	sealed flake path; feeder-to-barrel interface; drawing tolerance=OD44; ID29.20 +0.20/0; 4xØ4.60 +0.10/0 PCD36±0.10; compress to0.35–0.40; one installed plus one replacement; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FD-GSK-01.pdf	docs/final/complete_build_manual_ko.pdf step 10; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FD-HOP-01	sealed flake path; feeder-to-barrel interface; drawing tolerance=OD156 x straight145 + cone55; outlet ID24.90–25.00; lower flange OD44 x3, 4xØ4.60 +0.10/0 PCD36±0.05; register ID28.90 +0.03/0 x1.40±0.05; leak test; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FD-HOP-01.pdf	docs/final/complete_build_manual_ko.pdf step 9; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FD-MET-01	sealed flake path; feeder-to-barrel interface; drawing tolerance=OD29/ID25.00 +0.05/0 x105 flange face plus upper spigot OD28.77–28.80 x1.20±0.05; flanges Ø44 x3; 4xØ4.60 +0.10/0 PCD36±0.05; auger radial clearance 0.20–0.25; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FD-MET-01.pdf	docs/final/complete_build_manual_ko.pdf step 10; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FD-MET-02	sealed flake path; feeder-to-barrel interface; drawing tolerance=OD24.60 -0.05/0 x105; root Ø10 with local Ø12 x8 pin boss at Z4–12; bore Ø8.20 +0.10/0; Ø3.00 +0.05/0 cross-hole at Z8; pitch18; deburr/polish Ra≤3.2; Gate-2 sets mass/rev; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FD-MET-02.pdf	docs/final/complete_build_manual_ko.pdf step 10; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FD-MET-03	sealed flake path; feeder-to-barrel interface; drawing tolerance=Ø8 h8=7.978–8.000 x300; Ø3.00 +0.05/0 cross-holes at Z8 and Z292; lower hole matched with FD-MET-02, upper with FD-CP-01; paddle sweep Ø50/100/120 at Z183/228/273; straightness0.10; 2.2 N·m design torque SF≥2; datum=A=common journal axis; B=drive-end shoulder/face; C=keyway or retaining feature plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FD-MET-03.pdf	docs/final/complete_build_manual_ko.pdf step 10; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FH-01	sealed flake path; feeder feedback and cleanability	docs/drawings/v0.8/FD-003_positive_feeder.svg	docs/final/complete_build_manual_ko.pdf step 10; docs/final/assembly_steps.csv	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=DESIGNED; quantity_unit=each; not a primary dryer; v0.6.2.1 virtual feed envelope is active architecture; physical coefficient remains unverified
FH-02	sealed flake path; feeder feedback and cleanability	docs/drawings/v0.8/FD-003_positive_feeder.svg	docs/final/complete_build_manual_ko.pdf step 10; docs/final/assembly_steps.csv	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=RECEIPT_TEST_HOLD; quantity_unit=each; external predry not replaced; one-element power/equilibrium test sets final 4-8 quantity; vendor pinned: AliExpress 24 V insulated aluminum-shell PTC 35x21x5 class
FH-03	sealed flake path; feeder feedback and cleanability	docs/drawings/v0.8/FD-003_positive_feeder.svg	docs/final/complete_build_manual_ko.pdf step 10; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=USER_APPROVAL_REQUIRED; quantity_unit=each; exact donor labels ratios voltage current shaft and sensor levels required before pin assignment or energization
FM-AX-01	strand alignment and bearing/shaft fit; drawing tolerance=Ø8 h6 x80; TIR0.03; two metal collars; driven spindle interface remains donor-specific; datum=A=common journal axis; B=drive-end shoulder/face; C=keyway or retaining feature plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FM-AX-01.pdf	docs/final/complete_build_manual_ko.pdf step 16; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FM-EB-01	strand alignment and bearing/shaft fit; drawing tolerance=OD16 g6 x10; flange Ø24 x3; bore Ø8.20 +0.05/0 offset1.00 ±0.02; 2x3.4x10 flange slots at radius9; install paired bushes at equal index within0.5"; set full-rotation unloaded roller gap1.60-1.90 then clamp 2xM3 per bush; datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FM-EB-01.pdf	docs/final/complete_build_manual_ko.pdf step 16; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FM-GA-01	strand alignment and bearing/shaft fit; drawing tolerance=Ø5 h6=4.992-5.000 x30; SKF625-2Z bore4.992-5.000; two E-clips/collars outside PPR-C08; bearing inner-ring clamp must not preload outer rings; no printed axle; datum=A=common journal axis; B=drive-end shoulder/face; C=keyway or retaining feature plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FM-GA-01.pdf	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FM-GC-01	strand alignment and bearing/shaft fit; drawing tolerance=Ø32 x1; bore Ø15.00 +0.10/0 gives outer-ring radial overlap0.446-0.500; 3xØ3.4 PCD26; flatness0.05; bearing face burr-free; install flush in FM-GR-01 recess;	exports/final/manufacturing/RFQ/FM-GC-01.pdf	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv

Part ID	핵심 인터페이스	도면	조립 단계	펌웨어 의존	비고
	datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report				
FM-GR-01	strand alignment and bearing/shaft fit; drawing tolerance=OD36 x20; 2x Ø32 x1.05 cap recesses; 2x Ø16 H7 x5.10 bearing pockets from cap inner faces; Ø12 through relief; 3xØ3.4 PCD26 through; seat shoulders square.05; polished Ra<=1.6; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FM-GR-01.pdf	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FM-PL-01	strand alignment and bearing/shaft fit; drawing tolerance=100 x10 x40; fixed axle Ø8.2 at X29.10/Z20; adjustable bush seat Ø16 H7 at X69.90/Z20; 2xM3-6H blind7 at X69.90/Z11/29 from marked outer face; installed axes X29.10/X70.90 give gap1.80 and nip X50.00 local; 4xØ4.5 guard mounts; matched pair axis position ±0.05; datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FM-PL-01.pdf	docs/final/complete_build_manual_ko.pdf step 16; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FM-RL-01	strand alignment and bearing/shaft fit; drawing tolerance=finished OD40.00 ±0.025 x60; bore Ø8.2; TIR <=0.05; Shore A 50-70 sleeve; matched OD within 0.05; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/FM-RL-01.pdf	docs/final/complete_build_manual_ko.pdf step 16; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
FR-01	profile joint squareness and table load path	docs/drawings/v0.8/FR-001_frame.svg	docs/final/complete_build_manual_ko.pdf step 2; docs/final/assembly_steps.csv	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=project lab; cash_class=REUSE_PENDING; status=UNVERIFIED; quantity_unit=each; revision safety-orchestration-closure-v0.6.1; geometry unchanged from 60ccd92fe9a7df35b550a2a57649b1263da09d10; 14.668 m total; local 2040 shredder rails; exports/fabrication/frame_cut_list.csv
GD-01	moving/hot hazard reach protection and service interlock	docs/drawings/v0.8/GD-001_guards_panels.svg	docs/final/complete_build_manual_ko.pdf step 18; docs/final/assembly_steps.csv	hardware safety function; firmware monitoring only; never sole protection	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=DESIGNED; quantity_unit=lot; reach and fragment coupon pending
GGM_Chain Guard	기본 outer118 x24 x173, t1.0; 아래 전개 설명과 좌표 cutout을 따른다. STEP은 완성 외형.; 20 x10 x2 foot +20 x20 x2 upright tab2개를 용접. tab holeD3.40, front bearing plate에 M3x0.5.; 전면 개구는 두 bearing판의 실물 외곽으로 막혀야 한다. panel/plate gap0.30 nominal, 파편-손 접근 경로 확인.; 절곡 반경1.0, 각도90 +/-1deg. flat blank는 실제 절곡장비의 bend allowance 로 계산; 외곽치수는 완성치수.; 전면 mounting2개 외 service cover 잠금과 NC interlock은 전기 안전회로에 연결. 현장 무단 제거 금지.; GGM delta: 기본 outer118 x24 x173, t1.0; 아래 전개 설명과 좌표 cutout을 따른다. STEP은 완성 외형.; 20 x10 x2 foot +20 x20 x2 upright tab2개를 용접. tab holeD3.40, front bearing plate에 M3x0.5.; 전면 개구는 두 bearing판의 실물 외곽으로 막혀야 한다. panel/plate gap0.30 nominal, 파편-손 접근 경로 확인.; 절곡 반경1.0, 각도90 +/-1deg. flat blank는 실제 절곡 장비의 bend allowance로 계산; 외곽치수는 완성치수.; 전면 mounting2개 외 service cover 잠금과 NC interlock은 전기 안전회로에 연결. 현장 무단 제거 금지.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 8; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_EX_CouplingGuard	shell OD50/ID48, 길이58.00 +/-0.30. R2에서 실제 t1에 맞춰 재료 표기를 설정.; 상하2개 half-cover, seam gap1.00 +/-0.20. front flange OD76/ID48 x2, 각 half에2개 M3 clearance.; 장착4개 좌표는 조립도 참조. flange와 shell은 연속 접합 후 버 제거. 통전 전 service interlock 실물 확인.; GGM delta: shell OD50/ID48, 길이58.00 +/-0.30. R2에서 실제 t1에 맞춰 재료 표기를 설정.; 상하2개 half-cover, seam gap1.00 +/-0.20. front flange OD76/ID48 x2, 각 half에2개 M3 clearance.; 장착4개 좌표는 조립도 참조. flange와 shell은 연속 접합 후 버 제거. 통전 전 service interlock 실물 확인.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_EX_Fuse Input	S45C. 축방향 원점 Z0=입력 끝면, A=축 보어. Z0~22 OD28; Z22~26 OD46; Z26~30 OD20.; 보어 D12.050 +0.010/0 관통; 키홈 폭4.10 +0.03/0, 중심에서 홈 전장8.025. 외경/보어 TIR0.03.; 정합 pilot OD20.000 -0.020/0, Ra1.6. 정합부 축방향면 직각도 0.03.; 전단핀 hole D3.10 +0.02/0, R16.00 +/-0.03. Z22부터 flange 관통; 출력허브와 조립 jig에서 match-drill.; Z12의 radial D3.30은 M4x0.7 텡드릴; 유효나사5 mm 이상. set screw는 축방향 유지, 토크는 키로 전달.; GGM delta: S45C. 축방향 원점 Z0=입력 끝면, A=축 보어. Z0~22 OD28; Z22~26 OD46; Z26~30 OD20.; 보어 D12.050 +0.010/0 관통; 키홈 폭4.10 +0.03/0, 중심에서 홈 전장8.025. 외경/보어 TIR0.03.; 정합 pilot OD20.000 -0.020/0, Ra1.6. 정합부 축방향면 직각도 0.03.; 전단핀 hole D3.10 +0.02/0, R16.00 +/-0.03. Z22부터 flange 관통; 출력허브와 조립 jig에서 match-drill.; Z12의 radial D3.30은 M4x0.7 텡드릴; 유효나사5 mm 이상. set screw는 축방향 유지, 토크는 키로 전달.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_EX_Fuse Output	S45C. 공통 결합 datum Z26.40~30.40 OD46, Z30.40~52.40 OD28; 전장26.00 +/-0.05.; 보어 D12.050 +0.010/0, key4.10 +0.03/0. 입력면 counterbore D20.100 +0.020/0, 깊이4.20 +/-0.05.; 전단핀 D3.10 +0.02/0, R16.00 +/-0.03. 입력허브와 match-drill. 양 flange 사이 축간극0.40 +/-0.10.; 공통 datum Z40.40 radial D3.30은 M4x0.7, 유효나사5 mm 이상. 외경/보어 TIR0.03.; 핀 파단 뒤 pilot이 걸리지 않도록 무하중 자유회전 확인. 이 구조는 압력 방출장치가 아니다.; GGM delta: S45C. 공통 결합 datum Z26.40~30.40 OD46, Z30.40~52.40 OD28; 전장26.00 +/-0.05.; 보어 D12.050 +0.010/0, key4.10 +0.03/0. 입력면 counterbore D20.100 +0.020/0, 깊이4.20 +/-0.05.; 전단핀 D3.10 +0.02/0, R16.00 +/-0.03. 입력허브와 match-drill. 양 flange 사이 축간극0.40 +/-0.10.; 공통 datum Z40.40 radial D3.30은 M4x0.7, 유효나사5 mm 이상. 외경/보어 TIR0.03.; 핀 파	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest

Part ID	핵심 인터페이스	도면	조립 단계	펌웨어 의존	비고
GGM_EX_Fuse PinBlank	이 도면은 고정용 blank이며 운전용 전단핀 완성도가 아니다. 황동 lot를 구분한다.; 몸통 D3.00 -0.02/0 x11.70, 머리 D4.00 x1.00, 총12.70 +/-0.05.; 공통 coupling datum: 머리 Z20.70~21.70, 몸통21.70~33.40. 유지용 교차공 D0.90, 중심 Z32.00.; R16 단일 전 단 가정에서 목경 약2.416~2.483 mm는 계산 참고값만 해당. 목경을 확정값으로 가공 발주하지 않는다.; 설계검증 종료와 별도 시험 승인 뒤, 실제 결합 coupon의 8.8~9.3 N.m 해제 결과로 목경/길이를 확정한다.; GGM delta: 이 도면은 고정용 blank이며 운전용 전 단핀 완성도가 아니다. 황동 lot를 구분한다.; 몸통 D3.00 -0.02/0 x11.70, 머리 D4.00 x1.00, 총12.70 +/-0.05.; 공통 coupling datum: 머리 Z20.70~21.70, 몸통21.70~33.40. 유지용 교차공 D0.90, 중심 Z32.00.; R16 단일 전단 가정에서 목경 약 2.416~2.483 mm는 계산 참고값만 해당. 목경을 확정값으로 가공 발주하지 않는다.; 설계검증 종료와 별도 시험 승인 뒤, 실제 결합 coupon의 8.8~9.3 N.m 해제 결과로 목경/길이를 확정한다.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_EX_Motor Key	4x4: 길이25(2개),22(1),15(1),17(1). 6x6: 길이20(1개). 표의 bbox는 회전 배치 크기일 수 있다.; 4 mm 폭3.985~4.000, 6 mm 폭5.985~6.000; 높이는 nominal-0.03/0, 길이-0.20/0.; 키홈 천창과 키 상면0.10 이상 여유. 버 제거 C0.1, 억지 타격 금지. 실제 flank contact와 손회전 확인.; GGM delta: 4x4: 길이25(2개),22(1),15(1),17(1). 6x6: 길이20(1개). 표의 bbox는 회전 배치 크기일 수 있다.; 4 mm 폭3.985~4.000, 6 mm 폭5.985~6.000; 높이는 nominal-0.03/0, 길이-0.20/0.; 키홈 천창과 키 상면0.10 이상 여유. 버 제거 C0.1, 억지 타격 금지. 실제 flank contact와 손회전 확인.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_EX_Mount	S275 t6.00 +/-0.10. 120 x140 외곽 +/-0.20.; D36.50 +0.10/0, 4xD6.60 PCD104. 출력축 중심과 bolt-grid 중심은18.00 mm 다름.; 하부 2xD5.50 관통은 rail 체결. 도면 좌표를 따르고 기어박스 사각 중앙을 축으로 삼지 않는다.; A=모터 접촉면, B=출력축, C=바닥 모서리. 접촉면 평면도0.15.; R2: coupling cover 4xD2.50은 M3x0.5 관통탭, 유효5 mm 이상. 나사 끝단 돌출/모터 간섭 금지.; GGM delta: S275 t6.00 +/-0.10. 120 x140 외곽 +/-0.20.; D36.50 +0.10/0, 4xD6.60 PCD104. 출력축 중심과 bolt-grid 중심은18.00 mm 다름.; 하부 2xD5.50 관통은 rail 체결. 도면 좌표를 따르고 기어박스 사각 중앙을 축으로 삼지 않는다.; A=모터 접촉면, B=출력축, C=바닥 모서리. 접촉면 평면도0.15.; R2: coupling cover 4xD2.50은 M3x0.5 관통탭, 유효5 mm 이상. 나사 끝단 돌출/모터 간섭 금지.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_EX_Radial Cap	R2 변경: ID18에서 ID27.40 +0.10/0으로 수정. OD46.00 +/-0.10, t2.00 +/-0.05.; 2xD3.40 관통 PCD40. M3x10와 washer. 외륜만 유지하며 씰/내륜 접촉 금지.; 캡과 홀더의 선택 shim으로 axial endplay0.05~0.20. 강제로 조여 bearing을 pre-load하지 않는다.; GGM delta: R2 변경: ID18에서 ID27.40 +0.10/0으로 수정. OD46.00 +/-0.10, t2.00 +/-0.05.; 2xD3.40 관통 PCD40. M3x10와 washer. 외륜만 유지하며 씰/내륜 접촉 금지.; 캡과 홀더의 선택 shim으로 axial endplay0.05~0.20. 강제로 조여 bearing을 pre-load하지 않는다.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_EX_Radial Holder	S275 48 x48 x10.00 +/-0.05. 중심 D32.030 +/-0.010, Ra1.6.; 4xD4.50 관통, 38 x38 정사각. thrust plate의 대응 4개 M4x0.7과 결합.; 2xD2.50은 M3x0.5 캡용 관통탭, 유효6 mm 이상. 홀더 보어는 screw와 추력좌면 기준으로 정렬.; 베어링 정면을 압출 전달면으로 사용하지 않는다. 지정된51102 경로를 유지.; GGM delta: S275 48 x48 x10.00 +/-0.05. 중심 D32.030 +/-0.010, Ra1.6.; 4xD4.50 관통, 38 x38 정사각. thrust plate의 대응 4개 M4x0.7과 결합.; 2xD2.50은 M3x0.5 캡용 관통탭, 유효6 mm 이상. 홀더 보어는 screw와 추력좌면 기준으로 정렬.; 베어링 정면을 압출 추력 전달면으로 사용하지 않는다. 지정된51102 경로를 유지.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_EX_Screw Key	4x4: 길이25(2개),22(1),15(1),17(1). 6x6: 길이20(1개). 표의 bbox는 회전 배치 크기일 수 있다.; 4 mm 폭3.985~4.000, 6 mm 폭5.985~6.000; 높이는 nominal-0.03/0, 길이-0.20/0.; 키홈 천창과 키 상면0.10 이상 여유. 버 제거 C0.1, 억지 타격 금지. 실제 flank contact와 손회전 확인.; GGM delta: 4x4: 길이25(2개),22(1),15(1),17(1). 6x6: 길이20(1개). 표의 bbox는 회전 배치 크기일 수 있다.; 4 mm 폭3.985~4.000, 6 mm 폭5.985~6.000; 높이는 nominal-0.03/0, 길이-0.20/0.; 키홈 천창과 키 상면0.10 이상 여유. 버 제거 C0.1, 억지 타격 금지. 실제 flank contact와 손회전 확인.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 12; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_JackInner Spacer_front	모두 OD17.00 -0.05/0, ID12.10 +0.03/0. 길이는 대응 부품표 2.60/2.00/3.00/1.00 mm.; 길이 +/-0.02, 양 끝면 평행도0.02, Ra1.6. 외륜/씰이 아닌 내륜만 누른다.; 최종 collar와 shim은 두 bearing 사이 강제 예압 없이 축방향 유격0.05~0.20에서 선택.; GGM delta: 모두 OD17.00 -0.05/0, ID12.10 +0.03/0. 길이는 대응 부품표2.60/2.00/3.00/1.00 mm.; 길이 +/-0.02, 양 끝면 평행도0.02, Ra1.6. 외륜/씰이 아닌 내륜만 누른다.; 최종 collar와 shim은 두 bearing 사이 강제 예압 없이 축방향 유격0.05~0.20에서 선택.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_JackInner Spacer_pinionfront	모두 OD17.00 -0.05/0, ID12.10 +0.03/0. 길이는 대응 부품표 2.60/2.00/3.00/1.00 mm.; 길이 +/-0.02, 양 끝면 평행도0.02, Ra1.6. 외륜/씰이 아닌 내륜만 누른다.; 최종 collar와 shim은 두 bearing 사이 강제 예압 없이 축방향 유격0.05~0.20에서 선택.; GGM delta: 모두 OD17.00 -0.05/0, ID12.10 +0.03/0. 길이는 대응 부품표2.60/2.00/3.00/1.00 mm.; 길이 +/-0.02, 양 끝면 평행도0.02, Ra1.6. 외륜/씰이 아닌 내륜만 누른다.; 최종 collar와 shim은 두 bearing 사이 강제 예압 없이 축방향 유격0.05~0.20에서 선택.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest

Part ID	핵심 인터페이스	도면	조립 단계	평웨이 의존	비고
GGM_JackInner Spacer_pinionrear	모두 OD17.00 -0.05/0, ID12.10 +0.03/0. 길이는 대응 부품표 2.60/2.00/3.00/1.00 mm.; 길이 +/-0.02, 양 끝면 평행도0.02, Ra1.6. 외륜/썰이 아닌 내륜만 누른다.; 최종 collar와 shim은 두 bearing 사이 강제 예압 없이 축방향 유격0.05~0.20에서 선택.; GGM delta: 모두 OD17.00 -0.05/0, ID12.10 +0.03/0. 길이는 대응 부품표2.60/2.00/3.00/1.00 mm.; 길이 +/-0.02, 양 끝면 평행도0.02, Ra1.6. 외륜/썰이 아닌 내륜만 누른다.; 최종 collar와 shim은 두 bearing 사이 강제 예압 없이 축방향 유격0.05~0.20에서 선택.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_JackInner Spacer_rear	모두 OD17.00 -0.05/0, ID12.10 +0.03/0. 길이는 대응 부품표 2.60/2.00/3.00/1.00 mm.; 길이 +/-0.02, 양 끝면 평행도0.02, Ra1.6. 외륜/썰이 아닌 내륜만 누른다.; 최종 collar와 shim은 두 bearing 사이 강제 예압 없이 축방향 유격0.05~0.20에서 선택.; GGM delta: 모두 OD17.00 -0.05/0, ID12.10 +0.03/0. 길이는 대응 부품표2.60/2.00/3.00/1.00 mm.; 길이 +/-0.02, 양 끝면 평행도0.02, Ra1.6. 외륜/썰이 아닌 내륜만 누른다.; 최종 collar와 shim은 두 bearing 사이 강제 예압 없이 축방향 유격0.05~0.20에서 선택.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_12T	keyed torque path; radial TIR <= 0.10 mm; assembled axial shift+U95 <= 0.20 mm; chain alignment <= 0.20/150 mm; received maker axial retention required	docs/drawings/v0.8/SH-004_chain_phase_gear.svg	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM torque/current protection calibration; sprocket retention is mechanical	CURRENT GGM PURCHASED SPROCKET SOURCE: control/ggm_drive_contract.json + analysis/drive_acceptance_v08/drive_component_register.csv; key carries torque; maker retention hardware is axial-only; DRV-02 is superseded
GGM_SH_30T	keyed torque path; radial TIR <= 0.10 mm; assembled axial shift+U95 <= 0.20 mm; chain alignment <= 0.20/150 mm; received maker axial retention required	docs/drawings/v0.8/SH-004_chain_phase_gear.svg	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM torque/current protection calibration; sprocket retention is mechanical	CURRENT GGM PURCHASED SPROCKET SOURCE: control/ggm_drive_contract.json + analysis/drive_acceptance_v08/drive_component_register.csv; key carries torque; maker retention hardware is axial-only; DRV-02 is superseded
GGM_SH_Angle201	각 22 x36 x6 foot와22 x100 x6 web. 조립 높이106.00 +/-0.20.; 재고 angle을 잘라도 동일한 두께/외형/접촉면이 필요. 모터판과 바 닥판에3 mm 연속 필릿, 비드 간섭 제거.; 용접 jig 상태에서 축 위치 확인. 냉각 후 평면도0.15 및 수직도0.10/100 확인.; GGM delta: 각 22 x36 x6 foot와22 x100 x6 web. 조립 높이106.00 +/-0.20.; 재고 angle을 잘라도 동일한 두께/외형/접촉면이 필요. 모터판과 바 닥판에3 mm 연속 필릿, 비드 간섭 제거.; 용접 jig 상태에서 축 위치 확인. 냉각 후 평면도0.15 및 수직도0.10/100 확인.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Angle83	각 22 x36 x6 foot와22 x100 x6 web. 조립 높이106.00 +/-0.20.; 재고 angle을 잘라도 동일한 두께/외형/접촉면이 필요. 모터판과 바 닥판에3 mm 연속 필릿, 비드 간섭 제거.; 용접 jig 상태에서 축 위치 확인. 냉각 후 평면도0.15 및 수직도0.10/100 확인.; GGM delta: 각 22 x36 x6 foot와22 x100 x6 web. 조립 높이106.00 +/-0.20.; 재고 angle을 잘라도 동일한 두께/외형/접촉면이 필요. 모터판과 바 닥판에3 mm 연속 필릿, 비드 간섭 제거.; 용접 jig 상태에서 축 위치 확인. 냉각 후 평면도0.15 및 수직도0.10/100 확인.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Base	S275, 235 x 253 x 6.00 mm. 두께 6.00 +/-0.10; 외곽 +/-0.20.; 상면 A, X 최소 모서리 B, Y 최소 모서리 C. 구멍 위치는 좌표표 기준 +/-0.10.; 4xD6.60 +0.10/0: M6 관통. 4xD5.50 +0.10/0: profile 끝단 M5 체결용.; 지지면 평면도 0.15, 버 제거 C0.2. 가공 후 두 rail 위 흔들림 확인.; GGM delta: S275, 235 x 253 x 6.00 mm. 두께 6.00 +/-0.10; 외곽 +/-0.20.; 상면 A, X 최소 모서리 B, Y 최소 모서리 C. 구멍 위치는 좌표표 기준 +/-0.10.; 4xD6.60 +0.10/0: M6 관통. 4xD5.50 +0.10/0: profile 끝단 M5 체결용.; 지지면 평면도 0.15, 버 제거 C0.2. 가공 후 두 rail 위 흔들림 확인.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_BearingCap271	OD46.00 +/-0.10, ID27.40 +0.10/0, t2.00 +/-0.05. 2xD3.40 관통, PCD40.; 외륜에만 접촉. 썰/내륜과 간섭하지 않는지 수형 bearing으로 확인.; M3x10 class8.8와 washer; 유효몰림6 mm 이상, 바닥 여유1 mm 이상. 외륜 endplay0.05~0.20을 shim으로 맞춘다.; GGM delta: OD46.00 +/-0.10, ID27.40 +0.10/0, t2.00 +/-0.05. 2xD3.40 관통, PCD40.; 외륜에만 접촉. 썰/내륜과 간섭하지 않는지 수형 bearing으로 확인.; M3x10 class8.8와 washer; 유효몰림6 mm 이상, 바닥 여유1 mm 이상. 외륜 endplay0.05~0.20을 shim으로 맞춘다.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_BearingCap283	OD46.00 +/-0.10, ID27.40 +0.10/0, t2.00 +/-0.05. 2xD3.40 관통, PCD40.; 외륜에만 접촉. 썰/내륜과 간섭하지 않는지 수형 bearing으로 확인.; M3x10 class8.8와 washer; 유효몰림6 mm 이상, 바닥 여유1 mm 이상. 외륜 endplay0.05~0.20을 shim으로 맞춘다.; GGM delta: OD46.00 +/-0.10, ID27.40 +0.10/0, t2.00 +/-0.05. 2xD3.40 관통, PCD40.; 외륜에만 접촉. 썰/내륜과 간섭하지 않는지 수형 bearing으로 확인.; M3x10 class8.8와 washer; 유효몰림6 mm 이상, 바닥 여유1 mm 이상. 외륜 endplay0.05~0.20을 shim으로 맞춘다.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_BearingCap301	OD46.00 +/-0.10, ID27.40 +0.10/0, t2.00 +/-0.05. 2xD3.40 관통, PCD40.; 외륜에만 접촉. 썰/내륜과 간섭하지 않는지 수형 bearing으로 확인.; M3x10 class8.8와 washer; 유효몰림6 mm 이상, 바닥 여유1 mm 이상. 외륜 endplay0.05~0.20을 shim으로 맞춘다.; GGM delta: OD46.00 +/-0.10, ID27.40 +0.10/0, t2.00 +/-0.05. 2xD3.40 관통, PCD40.; 외륜에만 접촉. 썰/내륜과 간섭하지 않는지 수형 bearing으로 확인.; M3x10 class8.8와 washer; 유효몰림6 mm 이상, 바닥 여유1 mm 이상. 외륜 endplay0.05~0.20을 shim으로 맞춘다.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_BearingCap313	OD46.00 +/-0.10, ID27.40 +0.10/0, t2.00 +/-0.05. 2xD3.40 관통, PCD40.; 외륜에만 접촉. 썰/내륜과 간섭하지 않는지 수형 bearing으로 확인.; M3x10 class8.8와 washer; 유효몰림6 mm 이상, 바닥 여유1 mm 이상. 외륜 endplay0.05~0.20을 shim으로 맞춘다.; GGM delta: OD46.00 +/-0.10, ID27.40 +0.10/0, t2.00 +/-0.05. 2xD3.40 관통, PCD40.; 외륜에만 접촉. 썰/내륜과 간섭하지 않는지 수형 bearing으로 확인.; M3x10 class8.8와 washer; 유효몰림6 mm 이상, 바닥 여유1 mm 이상. 외륜 endplay0.05~0.20을 shim으로 맞춘다.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Bearing Plate273	S275, 216 x60 x10.00 +/-0.05. D32.030 +/-0.010; bore Ra1.6 이하.; 193 x40 hole grid는 축에 대하여 비대칭이다. 좌표표로 가공. 4xD5.50 관통.; 2xD2.50은 M3x0.5 탭드릴 표기. 양면에서 유 효 나사6 mm 이상; 캡용 2개 관통탭.; 두 판은 동일 jig로 보링. 합	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest

Part ID	핵심 인터페이스	도면	조립 단계	펌웨어 의존	비고
	성 축심차0.03 이하 목표; 외륜 endplay는 선택 shim으로 0.05~0.20.; R2 front판에만 chain guard tab용2xM3x0.5를 좌 표표대로 추가한다.; GGM delta: S275, 216 x60 x10.00 +/-0.05. D32.030 +/-0.010; bore Ra1.6 이하.; 193 x40 hole grid는 축에 대하여 비대칭이다. 좌표표로 가공. 4xD5.50 관통.; 2xD2.50은 M3x0.5 탭드릴 표기. 양면에서 유효 나사6 mm 이상; 캡용 2개 관 통탭.; 두 판은 동일 jig로 보링. 합성 축심차0.03 이하 목표; 외륜 endplay는 선택 shim으로0.05~0.20.; R2 front판에만 chain guard tab용2xM3x0.5를 좌표표대로 추가한다.				
GGM_SH_Bearing Plate303	S275, 216 x60 x10.00 +/-0.05. D32.030 +/-0.010; bore Ra1.6 이하.; 193 x40 hole grid는 축에 대하여 비대칭이다. 좌표표로 가 공. 4xD5.50 관통.; 2xD2.50은 M3x0.5 탭드릴 표기. 양면에서 유 효 나사6 mm 이상; 캡용 2개 관통탭.; 두 판은 동일 jig로 보링. 합 성 축심차0.03 이하 목표; 외륜 endplay는 선택 shim으로 0.05~0.20.; GGM delta: S275, 216 x60 x10.00 +/-0.05. D32.030 +/-0.010; bore Ra1.6 이하.; 193 x40 hole grid는 축에 대하여 비대칭이다. 좌표표로 가공. 4xD5.50 관통.; 2xD2.50은 M3x0.5 탭드릴 표기. 양면에서 유효 나사6 mm 이상; 캡용 2개 관 통탭.; 두 판은 동일 jig로 보링. 합성 축심차0.03 이하 목표; 외륜 endplay는 선택 shim으로0.05~0.20.	exports/final/manufacturing/drive_ggm/PPR_GGM_ MANUFACTURING_DRAWINGS_r2.pdf; exports/final/ manufacturing/drive_ggm/PPR_GGM_ MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_CouplingGuard	shell OD54/ID50 x57.00 +/-0.30; steel t2.0, 상하2개 half-cover.; seam gap1.00 +/-0.20, front flangeOD76/ID50 x2.0. flange4개 D3.40 mounting.; 장착판 대응4개M3x0.5. captive M3x8 기준; 실제물림/끝면여유 확인. flange용접 후 중심 정렬.; 커 플링과 최소 radial gap1.0 확인. 실제 NC service interlock은 별 도 수령-설계검증 후 연결.; GGM delta: shell OD54/ID50 x57.00 +/-0.30; steel t2.0, 상하2개 half-cover.; seam gap1.00 +/- 0.20, front flangeOD76/ID50 x2.0. flange4개 D3.40 mounting.; 장착판 대응4개M3x0.5. captiveM3x8 기준; 실제물 림/끝면여유 확인. flange용접 후 중심 정렬.; 커플링과 최소 radial gap1.0 확인. 실제 NC service interlock은 별도 수령-설계검증 후 연결.	exports/final/manufacturing/drive_ggm/PPR_GGM_ MANUFACTURING_DRAWINGS_r2.pdf; exports/final/ manufacturing/drive_ggm/PPR_GGM_ MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 8; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Cutter Key	4x4: 길이25(2개),22(1),15(1),17(1). 6x6: 길이20(1개). 표의 bbox는 회전 배치 크기일 수 있다.; 4 mm 폭3.985~4.000, 6 mm 폭5.985~6.000; 높이는 nominal-0.03/0, 길이-0.20/0.; 키홈 천창 과 키 상면0.10 이상 여유. 버 제거 C0.1, 억지 타격 금지. 실제 flank contact와 손회전 확인.; GGM delta: 4x4: 길이25(2 개),22(1),15(1),17(1). 6x6: 길이20(1개). 표의 bbox는 회전 배치 크기일 수 있다.; 4 mm 폭3.985~4.000, 6 mm 폭5.985~6.000; 높이는 nominal-0.03/0, 길이-0.20/0.; 키홈 천창과 키 상면0.10 이상 여유. 버 제거 C0.1, 억지 타격 금지. 실제 flank contact와 손 회전 확인.	exports/final/manufacturing/drive_ggm/PPR_GGM_ MANUFACTURING_DRAWINGS_r2.pdf; exports/final/ manufacturing/drive_ggm/PPR_GGM_ MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Fuse Input	S45C. 축방향 원점 Z0=입력 끝면, A=축 보어. Z0~22 OD28; Z22~26 OD46; Z26~30 OD20.; 보어 D12.050 +0.010/0 관통; 키 홈 폭4.10 +0.03/0, 중심에서 홈 천장8.025. 외경/보어 TIR0.03.; 정합 pilot OD20.000 -0.020/0, Ra1.6. 정합부 축방향면 직각도 0.03.; 전단핀 hole D3.10 +0.02/0, R16.00 +/-0.03. Z22부터 flange 관통; 출력허브와 조립 jig에서 match-drill.; Z12의 radial D3.30은 M4x0.7 탭드릴; 유효나사5 mm 이상. set screw는 축방 향 유지, 토크는 키로 전달.; GGM delta: S45C. 축방향 원점 Z0=입 력 끝면, A=축 보어. Z0~22 OD28; Z22~26 OD46; Z26~30 OD20.; 보어 D12.050 +0.010/0 관통; 키홈 폭4.10 +0.03/0, 중심 에서 홈 천장8.025. 외경/보어 TIR0.03.; 정합 pilot OD20.000 - 0.020/0, Ra1.6. 정합부 축방향면 직각도0.03.; 전단핀 hole D3.10 +0.02/0, R16.00 +/-0.03. Z22부터 flange 관통; 출력허브와 조립 jig에서 match-drill.; Z12의 radial D3.30은 M4x0.7 탭드릴; 유효 나사5 mm 이상. set screw는 축방향 유지, 토크는 키로 전달.	exports/final/manufacturing/drive_ggm/PPR_GGM_ MANUFACTURING_DRAWINGS_r2.pdf; exports/final/ manufacturing/drive_ggm/PPR_GGM_ MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Fuse Output	S45C. 공통 결합 datum Z26.40~30.40 OD46, Z30.40~52.40 OD28; 전장26.00 +/-0.05.; 보어 D12.050 +0.010/0, key4.10 +0.03/0. 입력면 counterbore D20.100 +0.020/0, 깊이4.20 +/- 0.05.; 전단핀 D3.10 +0.02/0, R16.00 +/-0.03. 입력허브와 match-drill. 양 flange 사이 축간극0.40 +/-0.10.; 공통 datum Z40.40 radial D3.30은 M4x0.7, 유효나사5 mm 이상. 외경/보어 TIR0.03.; 핀 파단 뒤 pilot이 걸리지 않도록 무하중 자유회전 확인. 이 구조는 압력 방출장치가 아니다.; GGM delta: S45C. 공통 결합 datum Z26.40~30.40 OD46, Z30.40~52.40 OD28; 전장26.00 +/-0.05.; 보어 D12.050 +0.010/0, key4.10 +0.03/0. 입력면 counterbore D20.100 +0.020/0, 깊이4.20 +/-0.05.; 전단핀 D3.10 +0.02/0, R16.00 +/-0.03. 입력허브와 match-drill. 양 flange 사이 축간극0.40 +/-0.10.; 공통 datum Z40.40 radial D3.30은 M4x0.7, 유효나사5 mm 이상. 외경/보어 TIR0.03.; 핀 파 단 뒤 pilot이 걸리지 않도록 무하중 자유회전 확인. 이 구조는 압력 방출장치가 아니다.	exports/final/manufacturing/drive_ggm/PPR_GGM_ MANUFACTURING_DRAWINGS_r2.pdf; exports/final/ manufacturing/drive_ggm/PPR_GGM_ MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Fuse PinBlank	이 도면은 고정용 blank이며 운전용 전단핀 완성도가 아니다. 황동 lot를 구분한다.; 몸통 D3.00 -0.02/0 x11.70, 머리 D4.00 x1.00, 총12.70 +/-0.05.; 공통 coupling datum: 머리 Z20.70~21.70, 몸 통21.70~33.40. 유지용 교차공 D0.90, 중심 Z32.00.; R16 단일 전 단 가정에서 목경 약2.416~2.483 mm는 계산 참고값만 해당. 목경 을 확정값으로 가공 발주하지 않는다.; 설계검증 종료와 별도 시험 승인 뒤, 실제 결합 coupon의 8.8~9.3 N.m 해체 결과로 목경/길이를 확정한다.; GGM delta: 이 도면은 고정용 blank이며 운전용 전 단핀 완성도가 아니다. 황동 lot를 구분한다.; 몸통 D3.00 -0.02/0 x11.70, 머리 D4.00 x1.00, 총12.70 +/-0.05.; 공통 coupling datum: 머리 Z20.70~21.70, 몸통21.70~33.40. 유지용 교차공 D0.90, 중심 Z32.00.; R16 단일 전단 가정에서 목경 약 2.416~2.483 mm는 계산 참고값만 해당. 목경을 확정값으로 가공 발주하지 않는다.; 설계검증 종료와 별도 시험 승인 뒤, 실제 결합 coupon의 8.8~9.3 N.m 해체 결과로 목경/길이를 확정한다.	exports/final/manufacturing/drive_ggm/PPR_GGM_ MANUFACTURING_DRAWINGS_r2.pdf; exports/final/ manufacturing/drive_ggm/PPR_GGM_ MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest

Part ID	핵심 인터페이스	도면	조립 단계	펌웨어 의존	비고
GGM_SH_Jack InputKey	4x4: 길이25(2개),22(1),15(1),17(1). 6x6: 길이20(1개). 표의 bbox는 회전 배치 크기일 수 있다.; 4 mm 폭3.985~4.000, 6 mm 폭5.985~6.000; 높이는 nominal-0.03/0, 길이-0.20/0.; 키홀 천창과 키 상면0.10 이상 여유. 버 제거 C0.1, 억지 타격 금지. 실제 flank contact와 손회전 확인.; GGM delta: 4x4: 길이25(2개),22(1),15(1),17(1). 6x6: 길이20(1개). 표의 bbox는 회전 배치 크기일 수 있다.; 4 mm 폭3.985~4.000, 6 mm 폭5.985~6.000; 높이는 nominal-0.03/0, 길이-0.20/0.; 키홀 천창과 키 상면0.10 이상 여유. 버 제거 C0.1, 억지 타격 금지. 실제 flank contact와 손회전 확인.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Jack SprocketKey	4x4: 길이25(2개),22(1),15(1),17(1). 6x6: 길이20(1개). 표의 bbox는 회전 배치 크기일 수 있다.; 4 mm 폭3.985~4.000, 6 mm 폭5.985~6.000; 높이는 nominal-0.03/0, 길이-0.20/0.; 키홀 천창과 키 상면0.10 이상 여유. 버 제거 C0.1, 억지 타격 금지. 실제 flank contact와 손회전 확인.; GGM delta: 4x4: 길이25(2개),22(1),15(1),17(1). 6x6: 길이20(1개). 표의 bbox는 회전 배치 크기일 수 있다.; 4 mm 폭3.985~4.000, 6 mm 폭5.985~6.000; 높이는 nominal-0.03/0, 길이-0.20/0.; 키홀 천창과 키 상면0.10 이상 여유. 버 제거 C0.1, 억지 타격 금지. 실제 flank contact와 손회전 확인.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Jackshaft	S45C 공급 상태는 상온 항복강도300 MPa 이상 근거 필요. 전장 78.00 +/-0.10.; D12 h6=11.989~12.000. 베어링 land Ra0.8 이하, 기타 원통 Ra1.6 이하.; 축 원점=전방 끝면. 키홀1 Z2.40~24.40; 키홀2 Z39.00~57.00. 폭4.00 +/-0.02/0; 바닥은 축중심에서3.50 mm.; 베어링 land Z27~37 및57~67은 원통면을 보존. 끝면 C0.2, 키홀 뿌리 R0.2 이하.; 두 land 기준 TIR0.03 이하. h6는 현재 원형 설계 기준이며 수령 bearing과 실제 fit/creep 판정 전 운전 금지.; GGM delta: S45C 공급 상태는 상온 항복강도 300 MPa 이상 근거 필요. 전장78.00 +/-0.10.; D12 h6=11.989~12.000. 베어링 land Ra0.8 이하, 기타 원통 Ra1.6 이하.; 축 원점=전방 끝면. 키홀1 Z2.40~24.40; 키홀2 Z39.00~57.00. 폭4.00 +/-0.02/0; 바닥은 축중심에서3.50 mm.; 베어링 land Z27~37 및57~67은 원통면을 보존. 끝면 C0.2, 키홀 뿌리 R0.2 이하.; 두 land 기준 TIR0.03 이하. h6는 현재 원형 설계 기준이며 수령 bearing과 실제 fit/creep 판정 전 운전 금지.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Motor Key	4x4: 길이25(2개),22(1),15(1),17(1). 6x6: 길이20(1개). 표의 bbox는 회전 배치 크기일 수 있다.; 4 mm 폭3.985~4.000, 6 mm 폭5.985~6.000; 높이는 nominal-0.03/0, 길이-0.20/0.; 키홀 천창과 키 상면0.10 이상 여유. 버 제거 C0.1, 억지 타격 금지. 실제 flank contact와 손회전 확인.; GGM delta: 4x4: 길이25(2개),22(1),15(1),17(1). 6x6: 길이20(1개). 표의 bbox는 회전 배치 크기일 수 있다.; 4 mm 폭3.985~4.000, 6 mm 폭5.985~6.000; 높이는 nominal-0.03/0, 길이-0.20/0.; 키홀 천창과 키 상면0.10 이상 여유. 버 제거 C0.1, 억지 타격 금지. 실제 flank contact와 손회전 확인.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Mount	S275 t6.00 +/-0.10. 140 x 190.167, 외곽 +/-0.20.; D36.50 +/-0.10/0 통과공은 정밀 파일럿이 아니다. 축 정렬은 조립 검사로 확보.; 4xD6.60 +/-0.10/0, PCD104 중심은 출력축에서 18.00 mm 편심. 표를 따라 가공.; M6 관통 볼트 사용. 두 보강 angle을 jig에서 3 mm 연속 필릿 용접. 용접 뒤 정렬면 평면도0.15.; R2: coupling cover 4xD2.50은 M3x0.5 관통탭, 유효5 mm 이상. 나사 끝단 돌출/모터 간섭 금지.; GGM delta: S275 t6.00 +/-0.10. 140 x 190.167, 외곽 +/-0.20.; D36.50 +/-0.10/0 통과공은 정밀 파일럿이 아니다. 축 정렬은 조립 검사로 확보.; 4xD6.60 +/-0.10/0, PCD104 중심은 출력축에서 18.00 mm 편심. 표를 따라 가공.; M6 관통 볼트 사용. 두 보강 angle을 jig에서 3 mm 연속 필릿 용접. 용접 뒤 정렬면 평면도0.15.; R2: coupling cover 4xD2.50은 M3x0.5 관통탭, 유효5 mm 이상. 나사 끝단 돌출/모터 간섭 금지.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Spacer_220_100	OD12.00 +/-0.10, ID6.60 +/-0.10/0, nominal L10.00 +/-0.05.; 체인 장력은4개를 동일 길이8~13 mm로 조절. 같은 set의 길이 차이 0.03 이하.; 서로 다른 spacer 높이로 축 기울기를 만들지 않는다. 연질 출력 spacer 금지.; GGM delta: OD12.00 +/-0.10, ID6.60 +/-0.10/0, nominal L10.00 +/-0.05.; 체인 장력은4개를 동일 길이 8~13 mm로 조절. 같은 set의 길이 차이0.03 이하.; 서로 다른 spacer 높이로 축 기울기를 만들지 않는다. 연질 출력 spacer 금지.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Spacer_220_280	OD12.00 +/-0.10, ID6.60 +/-0.10/0, nominal L10.00 +/-0.05.; 체인 장력은4개를 동일 길이8~13 mm로 조절. 같은 set의 길이 차이 0.03 이하.; 서로 다른 spacer 높이로 축 기울기를 만들지 않는다. 연질 출력 spacer 금지.; GGM delta: OD12.00 +/-0.10, ID6.60 +/-0.10/0, nominal L10.00 +/-0.05.; 체인 장력은4개를 동일 길이 8~13 mm로 조절. 같은 set의 길이 차이0.03 이하.; 서로 다른 spacer 높이로 축 기울기를 만들지 않는다. 연질 출력 spacer 금지.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Spacer_90_100	OD12.00 +/-0.10, ID6.60 +/-0.10/0, nominal L10.00 +/-0.05.; 체인 장력은4개를 동일 길이8~13 mm로 조절. 같은 set의 길이 차이 0.03 이하.; 서로 다른 spacer 높이로 축 기울기를 만들지 않는다. 연질 출력 spacer 금지.; GGM delta: OD12.00 +/-0.10, ID6.60 +/-0.10/0, nominal L10.00 +/-0.05.; 체인 장력은4개를 동일 길이 8~13 mm로 조절. 같은 set의 길이 차이0.03 이하.; 서로 다른 spacer 높이로 축 기울기를 만들지 않는다. 연질 출력 spacer 금지.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
GGM_SH_Spacer_90_280	OD12.00 +/-0.10, ID6.60 +/-0.10/0, nominal L10.00 +/-0.05.; 체인 장력은4개를 동일 길이8~13 mm로 조절. 같은 set의 길이 차이 0.03 이하.; 서로 다른 spacer 높이로 축 기울기를 만들지 않는다. 연질 출력 spacer 금지.; GGM delta: OD12.00 +/-0.10, ID6.60 +/-0.10/0, nominal L10.00 +/-0.05.; 체인 장력은4개를 동일 길이 8~13 mm로 조절. 같은 set의 길이 차이0.03 이하.; 서로 다른 spacer 높이로 축 기울기를 만들지 않는다. 연질 출력 spacer 금지.	exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf; exports/final/manufacturing/drive_ggm/PPR_GGM_MANUFACTURING_DRAWINGS_r2.pdf	docs/final/complete_build_manual_ko.pdf step 7; docs/final/assembly_steps.csv	GGM/BTS7960 released profile; commissioning calibration remains NOT_RUN	authoritative sub-manifest: exports/final/manufacturing/drive_ggm/manifest.csv; GGM drive delta drawing bound by drive_ggm manifest
HP-01	anti-reach opening and removable hopper interface	docs/drawings/v0.8/FD-001_hopper.svg	docs/final/complete_build_manual_ko.pdf step 9; docs/final/assembly_steps.csv	NONE	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=DESIGNED; quantity_unit=each; 4.5 L usable; fabrication STEP/DXF/note under exports/fabrication

Part ID	핵심 인터페이스	도면	조립 단계	펌웨어 의존	비고
IN-HOP-01	anti-reach opening and removable hopper interface; drawing tolerance=OD200 x straight150 + cone60; outlet Ø40; wall 2.0; lid rail datum flatness 0.5; leak-free dry-flake seams; datum=A=material-flow axis; B=mounting face; C=drive/sensor centre plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/IN-HOP-01.pdf	docs/final/complete_build_manual_ko.pdf step 9; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
PL-01	roller alignment, pinch guard and tach calibration	docs/drawings/v0.8/FM-002_gauge_puller.svg	docs/final/complete_build_manual_ko.pdf step 16; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=UNVERIFIED; quantity_unit=each; closed inner speed PI; roller diameter and pulses/rev require CRC calibration; matched plates/rollers and metal spindles
PPR-C01	printed interface per dimension sheet; no structural hot/ high-load path; mating=metal hopper rails and lid-interlock flag; tolerance=finished latch hole Ø4.50–4.70; screw head flush or recessed≤0.05; rail slide gap0.35	exports/print/PPR-C01/dimension_sheet.svg	docs/final/complete_build_manual_ko.pdf step 9; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C02	printed interface per dimension sheet; no structural hot/ high-load path; mating=hopper and metal cutter chamber; tolerance=0.40 mm flake path	exports/print/PPR-C02/dimension_sheet.svg	docs/final/complete_build_manual_ko.pdf step 9; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C03	printed interface per dimension sheet; no structural hot/ high-load path; mating=1.00 ±0.05 mm sheet bin and screen rails; tolerance=sheet slot 1.40 ±0.30 mm	exports/print/PPR-C03/dimension_sheet.svg	docs/final/complete_build_manual_ko.pdf step 10; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C04	printed interface per dimension sheet; no structural hot/ high-load path; mating=metal screen; tolerance=0.25 mm	exports/print/PPR-C04/dimension_sheet.svg	docs/final/complete_build_manual_ko.pdf step 9; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C05	printed interface per dimension sheet; no structural hot/ high-load path; mating=80 mm fan and next duct; tolerance=0.30 mm flange registration	exports/print/PPR-C05/dimension_sheet.svg	docs/final/complete_build_manual_ko.pdf step 14; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C06	printed interface per dimension sheet; no structural hot/ high-load path; mating=LED/photodiode cross frame; tolerance=finished holes Ø3.40–3.50; 0.20 mm optical slit; install X/Y pair on one straight Ø1.75 calibration wire, match-drill mounting datum, scan both axes and accept optical centreline offset <=0.10 mm	exports/print/PPR-C06/dimension_sheet.svg	docs/final/complete_build_manual_ko.pdf step 15; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C07	printed interface per dimension sheet; no structural hot/ high-load path; mating=metal puller plate; tolerance=0.40 mm guard gap	exports/print/PPR-C07/dimension_sheet.svg	docs/final/complete_build_manual_ko.pdf step 16; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C08	printed interface per dimension sheet; no structural hot/ high-load path; mating=FM-GA-01 fixed Ø5 axle and profile; 625 bearings are seated in FM-GR-01; tolerance=finished bore Ø5.20–5.40; align both loose T-slot brackets on the same axle before 2.0 N.m torque; axle shall pass by hand without visible bending	exports/print/PPR-C08/dimension_sheet.svg	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C09	printed interface per dimension sheet; no structural hot/ high-load path; mating=12 mm metal spindle, metal collar, and received spool; tolerance=finished spindle bore Ø12.20–12.40; diametral clearance0.20–0.411 to Ø12 h6; cone-to-received-spool contact is separately selected	exports/print/PPR-C09/dimension_sheet.svg	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C10	printed interface per dimension sheet; no structural hot/ high-load path; mating=specified Ø8 h6 rods and GT2 belt; tolerance=finished rod bores Ø8.40–8.50; finished clamp holes Ø4.50–4.70; use two Ø8 h6=7.991–8.000 ground steel rods	exports/print/PPR-C10/dimension_sheet.svg	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C11	printed interface per dimension sheet; no structural hot/ high-load path; mating=metal control panel; tolerance=finished holes Ø3.40–3.50; 0.25 mm TFT	exports/print/PPR-C11/dimension_sheet.svg	docs/final/complete_build_manual_ko.pdf step 22; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-C12	printed interface per dimension sheet; no structural hot/ high-load path; mating=20 mm profile and fixed 18x18 cable duct; tolerance=18.6 mm cavity; 0.30 mm/side	exports/print/PPR-C12/dimension_sheet.svg	docs/final/complete_build_manual_ko.pdf step 20; docs/final/assembly_steps.csv	NONE	SUBORDINATE DETAIL SOURCE: exports/print/print_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
PPR-EXTRUDER-ASM	reference identity must resolve to canonical BOM/drawing	docs/drawings/v0.8/EX-001_extruder_assembly.svg	docs/final/complete_build_manual_ko.pdf step 11; docs/final/assembly_steps.csv	NONE	active_part_set reference; canonical=PPR-EXTRUDER-ASM; exclude from procurement roll-up
PPR-FEEDER-ASM	reference identity must resolve to canonical BOM/drawing	docs/drawings/v0.8/FD-003_positive_feeder.svg	docs/final/complete_build_manual_ko.pdf step 10; docs/final/assembly_steps.csv	NONE	active_part_set reference; canonical=PPR-FEEDER-ASM; exclude from procurement roll-up
PPR-FORMING-ASM	reference identity must resolve to canonical BOM/drawing	docs/drawings/v0.8/FM-001_cooling_strand_path.svg	docs/final/complete_build_manual_ko.pdf step 14; docs/final/assembly_steps.csv	NONE	active_part_set reference; canonical=PPR-FORMING-ASM; exclude from procurement roll-up
PPR-FRAME-ASM	reference identity must resolve to canonical BOM/drawing	docs/drawings/v0.8/FR-001_frame.svg	docs/final/complete_build_manual_ko.pdf step 2; docs/final/assembly_steps.csv	NONE	active_part_set reference; canonical=PPR-FRAME-ASM; exclude from procurement roll-up
PPR-FULL-ASM	reference identity must resolve to canonical BOM/drawing	docs/drawings/v0.8/ASM-001_full_assembly.svg	docs/final/complete_build_manual_ko.pdf step 1; docs/final/assembly_steps.csv	NONE	active_part_set reference; canonical=PPR-FULL-ASM; exclude from procurement roll-up
PPR-SHREDDER-ASM	reference identity must resolve to canonical BOM/drawing	docs/drawings/v0.8/SH-001_shredder_assembly.svg	docs/final/complete_build_manual_ko.pdf step 4; docs/final/assembly_steps.csv	NONE	active_part_set reference; canonical=PPR-SHREDDER-ASM; exclude from procurement roll-up
SF-01	hardwired E-stop/interlock/thermal chain independent of firmware	docs/drawings/v0.8/EL-001_electrical_enclosure.svg	docs/final/complete_build_manual_ko.pdf step 21; docs/final/assembly_steps.csv	hardware safety function; firmware monitoring only; never sole protection	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=stock/buy; cash_class=MIXED; status=PHYSICAL_GATE; quantity_unit=each; software independent
SF-02	hardwired E-stop/interlock/thermal chain independent of firmware	docs/drawings/v0.8/EL-001_electrical_enclosure.svg	docs/final/complete_build_manual_ko.pdf step 21; docs/final/assembly_steps.csv	hardware safety function; firmware monitoring only; never sole protection	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=stock/buy; cash_class=MIXED; status=PHYSICAL_GATE; quantity_unit=each; hardware branch inhibit
SF-03	hardwired E-stop/interlock/thermal chain independent of firmware	docs/drawings/v0.8/EL-001_electrical_enclosure.svg	docs/final/complete_build_manual_ko.pdf step 21; docs/final/assembly_steps.csv	hardware safety function; firmware monitoring only; never sole protection	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=stock/buy; cash_class=MIXED; status=PHYSICAL_GATE; quantity_unit=lot; not optional; TH-FUSE-01 quantity 3 = TF-BARREL + TF-DIE installed in series in K0 coil chain + one same-spec spare; F-H1..F-H4 are separate 5 A branch overcurrent fuses; exact MPN/rating per thermal cutoff contract and heater RFQ receipt gate
SH-01	guarded cutter torque path and service lockout	docs/drawings/v0.8/SH-001_shredder_assembly.svg	docs/final/complete_build_manual_ko.pdf step 8; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=CNC; cash_class=CASH; status=COUPON_HOLD; quantity_unit=each; common PLA/PET path
SH-02	guarded cutter torque path and service lockout	docs/drawings/v0.8/SH-001_shredder_assembly.svg	docs/final/complete_build_manual_ko.pdf step 8; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=CNC; cash_class=CASH; status=COUPON_HOLD; quantity_unit=each; metal shim clearance
SH-03	guarded cutter torque path and service lockout	docs/drawings/v0.8/SH-001_shredder_assembly.svg	docs/final/complete_build_manual_ko.pdf step 8; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=buy + project-lab fabrication; cash_class=MIXED; status=SELECTED_PURCHASE_HOLD; quantity_unit=each; selected shredder drive; gearbox output 40 rpm, cutter ~16 rpm through 2.5:1 chain; digital load path covers 19.6 N.m available and 8.0 N.m software gearbox limit; receipt/alignment/pin/current physical checks NOT_RUN
SH-04	guarded cutter torque path and service lockout	docs/drawings/v0.8/SH-001_shredder_assembly.svg	docs/final/complete_build_manual_ko.pdf step 8; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=DESIGNED; quantity_unit=each; GGM_SH_12T uses 4x4 key

Part ID	핵심 인터페이스	도면	조립 단계	펌웨어 의존	비고
					on jackshaft and GGM_SH_30T uses 6x6 key directly on CUT-05R; keys carry torque while received maker retention hardware is axial-only and receipt-gated; radial TIR <=0.10 mm, chain alignment <=0.20/150 mm, axial shift+U95 <=0.20 mm; DRV-02 superseded; GGM mechanical protection target 8.8-9.3 N.m at gearbox-side coupling
SH-05	guarded cutter torque path and service lockout	docs/drawings/v0.8/SH-001_shredder_assembly.svg	docs/final/complete_build_manual_ko.pdf step 8; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=project lab + sensor stock-check/buy; cash_class=MIXED; status=REUSE_CONFIRMED_SENSOR_HOLD; quantity_unit=each; BTS7960 qty1 of owned qty2 allocated to shredder; D5 RPWM/D4 LPWM/D32 enable; 6 A software overcurrent and 8.0 N.m calibrated gearbox torque limit; current sensor/calibration NOT_RUN
SP-01	spindle/bearing fit; dancer/traverse envelope	docs/drawings/v0.8/SP-001_spooler_traverse.svg	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	AUTHORITATIVE DESIGN SOURCE: bom/bom.csv; source=mixed; cash_class=MIXED; status=DESIGNED; quantity_unit=each; 1 kg spool; A15 spool tach and A5/A6 limits; radius remains geometry/wound-length estimate with uncertainty
SP-AX-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=Ø8 h6 =7.991-8.000 x28; metal collars; one pivot and one end roller axle; datum=A=common journal axis; B=drive-end shoulder/face; C=keyway or retaining feature plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-AX-01.pdf	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-BP-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=105 x10 x60; bearing centre X30/Z30; Ø28.000–28.021 H7 x8.05–8.10 pocket from marked inner face; Ø26 through relief leaves 1.95–2.00 shoulder; 4xØ5.5 retainer + 2xØ5.5 profile tab; matched axis position ±0.05; datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-BP-01.pdf	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-BR-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=54 x2 x54; Ø26.0 +0.10/0 relief gives 0.95–1.00 radial outer-ring overlap; 4xØ5.5 at 44 mm square; flatness0.10; burr away from bearing; use 4x M5 with SP-BP-01; datum=A=frame mounting face; B=primary bearing axis; C=secondary bearing/fastener axis; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-BR-01.pdf	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-DA-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=100 mm pivot centres; 12 mm arm; 2xØ8.20 +0.05/0; edge R2; full -25..+25deg motion; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-DA-01.pdf	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-DS-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=36 x8 x80; pivot Ø8.20 +0.05/0 at X18/Z45; 2xØ5.5 foot mounts; metal support carries spring/tension load; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-DS-01.pdf	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-MM-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=101 x6 x52; motor centre X26, Ø24; 4xØ4.5 at 31 mm square; 2xØ5.5 profile tab; actual donor shaft and body measurement required before coupling release; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-MM-01.pdf	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-RL-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=OD20 x20; bore Ø8.20 +0.05/0; diametral axle clearance0.20-0.259; free rotation under 0.2-1.0 N filament tension; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-RL-01.pdf	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-SH-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=Ø12 h6=11.989-12.000 x143; straightness0.05; two SKF6001-2RSH bore11.992-12.000; axial collars carry spool load; datum=A=common journal axis; B=drive-end shoulder/face; C=keyway or retaining feature plane; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-SH-01.pdf	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
SP-TR-01	spindle/bearing fit; dancer/traverse envelope; drawing tolerance=5 x50 x40; two finished Ø8.20–8.30 at 25 mm spacing; 2xØ5.5 mount; use two Ø8 h6=7.991–8.000 x160 ground steel rods; match-ream pair and accept rod parallelism <=0.10/160; datum=A=frame mounting plane; B=primary edge; C=mounting-hole centreline; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/SP-TR-01.pdf	docs/final/complete_build_manual_ko.pdf step 17; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/fabrication/machine_manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
TH-BH-01	heater fit, insulation, branch fuse and independent thermal cutoff; drawing tolerance=24 VDC 100 W each; free-state ID34.10–34.20; usable split-closure travel >=1.00; width45 ±0.5; radial build2 nominal; cold resistance 5.76 Ω ±10%; 300 mm fiberglass leads; PE-bonded sheath; PET service 300 C design; no stock Ø35 substitution; datum=A=profile mounting plane; B=barrel axis; C=rear axial datum; verify datum marking on supplier inspection report	exports/final/manufacturing/RFQ/TH-BH-01.pdf	docs/final/complete_build_manual_ko.pdf step 13; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/thermal/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval; FINAL MANUFACTURING DETAIL: exports/final/manufacturing/RFQ/manifest.csv
TH-DIE-01	heater fit, insulation, branch fuse and independent thermal cutoff	exports/thermal/parts/TH-DIE-01/drawing_notes.md	docs/final/complete_build_manual_ko.pdf step 13; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/thermal/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
TH-FUSE-01	heater fit, insulation, branch fuse and independent thermal cutoff	exports/thermal/parts/TH-FUSE-01/drawing_notes.md	docs/final/complete_build_manual_ko.pdf step 13; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/thermal/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
TH-TC-01	heater fit, insulation, branch fuse and independent thermal cutoff	exports/thermal/parts/TH-TC-01/drawing_notes.md	docs/final/complete_build_manual_ko.pdf step 13; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/thermal/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval
TH-TCR-01	heater fit, insulation, branch fuse and independent thermal cutoff	exports/thermal/parts/TH-TCR-01/drawing_notes.md	docs/final/complete_build_manual_ko.pdf step 13; docs/final/assembly_steps.csv	firmware/arduino_mega/src + released pin map/ calibration; physical calibration required	SUBORDINATE DETAIL SOURCE: exports/thermal/manifest.csv; parent design authority remains bom/bom.csv; procurement/fabrication requires user approval

4 일정표 요약

Fastener 30행 · Consumables 5행 · Tools 13행 · Alternatives 12행.

4.1 미검증 체결품 — 체결/조립 진행 보류

Joint	후보 규격	토크 상태	해소 필요 조건
SYS-09	received sprocket maker axial-retention hardware; 4x4/6x6 keys carry torque	HOLD	blue-check key flank; no friction-only/set-screw-only torque path; each sprocket radial TIR <=0.10 mm; total axial shift+U95 <=0.20 mm; chain alignment <=0.20/150 mm
SYS-12	M4x16 A2-70 SHCS + washer + all-metal prevailing nut	HOLD	register clearance0.10-0.16; tighten crosswise only until gasket thickness0.35-0.40; dry-flake leak/retention test required

세부값은 동봉 CSV와 BOM.xlsx의 동일 이름 sheet가 지배한다. USER_APPROVAL_REQUIRED, RECEIPT_TEST_REQUIRED, NOT_APPROVED는 누락이 아니라 의도적인 물리/조달 gate이며 승인으로 승격하지 않는다.